



ONTAP FlexGroup Volume Advanced

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1 Introduction

This lab demonstrates a number of the more advanced features of FlexGroup volumes, and provides deeper insight into FlexGroup volume operation. It builds on foundational concepts introduced during the ONTAP FlexGroup Volumes Fundamentals lab. If you have not yet taken that lab, we recommend that you do so before starting this one.

What is a NetApp FlexGroup?

Recall from the ONTAP FlexGroup Volumes Fundamentals lab that a NetApp FlexGroup volume is a single volume namespace comprised of multiple constituent volumes that together appear to storage users like any other FlexVol volume. FlexGroup volumes provide greatly increased capacity and performance over FlexVol volumes without requiring storage users to alter how they access and use ONTAP NAS storage. Many aspects of FlexGroup administration are very similar to that of FlexVols, so storage administrators can adopt FlexGroup volumes without significant re-training or re-tooling.

A NetApp FlexGroup blends the following capabilities:

- **Capacity**—NetApp FlexGroup volumes provide massive capacity and file counts (20PB, 400 billion files) by using multiple FlexVol member volumes together to present a single, unified namespace across multiple aggregates and nodes in a cluster.
- **Nondisruptive Scale**—With NetApp FlexGroup, a single namespace can live on multiple aggregates on one or more nodes in a cluster, and can expand to new nodes or aggregates with no downtime so that you can nondisruptively add performance or capacity to an existing workload.
- **Performance**—A NetApp FlexGroup volume does not just provide immense capacity by virtue of using multiple constituent member volumes, it also offers vastly improved performance for high metadata workloads by allowing file ingest to occur in parallel across multiple volume affinities. This not only enables the most efficient use of your storage hardware, it also prevents bottlenecks during high ingest workloads.
- **Simplicity**—NetApp FlexGroup volumes provide massive storage capacity and support enormous file counts. They transparently scale across cluster nodes during volume creation, they automatically load balance new file and folder placement across member volumes during volume operation, and they are even capable of automatically re-sizing themselves as they approach capacity thresholds. By self-managing these details, FlexGroups free storage administrators and users to worry less about storage and instead focus on more productive activities.

2 Lab Activities

This lab has the following activities:

- [FlexGroup File Layout and Balance](#)
- [Exploring FlexGroup Autosize Behavior](#)
- [Add New Constituent Volumes to an Existing FlexGroup](#)
- [Automatically Rebalance a FlexGroup Volume](#)
- [FlexGroup Volume Move](#)
- [Use SnapMirror to Restore a Single File](#)
- [Convert a FlexVol to a FlexGroup](#)

2.1 FlexGroup File Layout and Balance

ONTAP works during file ingestion to distribute FlexGroup volume workloads evenly across constituent (aka member) volumes to maintain a balance of constituent volume capacity and file counts. In this lab activity, a storage administrator wants to understand how ONTAP distributes files and their storage space consumption across FlexGroup constituent volumes during initial file ingestion, specifically for the “largeFG” FlexGroup volume.

The “largeFG” FlexGroup volume’s lineage traces back to the work flow from the “ONTAP FlexGroup Fundamentals” lab. As a reminder, this FlexGroup volume is comprised of four constituent volumes, each approximately 150 GB in size, for a total FlexGroup volume capacity of 600 GB. The “largeFG” FlexGroup volume contains a dataset of over 100,000 files and directories whose state after initial creation is captured in a volume snapshot named “base”. Finally, the “largeFG” FlexGroup volume also has an established data protection relationship to the “vol_largeFG_dest” FlexGroup volume on cluster2.

1. This lab utilizes in-lab automation as a part of the lab provisioning process. Verify that your lab session’s in-lab automation completed successfully by examining the desktop of Jumphost. If you see a green checkmark shortcut entitled “Automation completed” then the provisioning completed successfully and you can proceed to the next step in this guide.



Tip: The automation often completes prior to you connecting to your lab session, but that is not a guaranteed outcome. If you don’t see the green checkmark appear within eight minutes of connecting to your lab session, or if the automation fails, then please go to the portal you used to connect to the lab and click the “Help” button found there (located on the same row as the “Join” button) to contact the lab support team for assistance.



Figure 2-1:

2. On the task bar of Jumphost, launch **PuTTY**.



Figure 2-2:

3. Double-click **rhel1** in the “Saved Sessions” list.

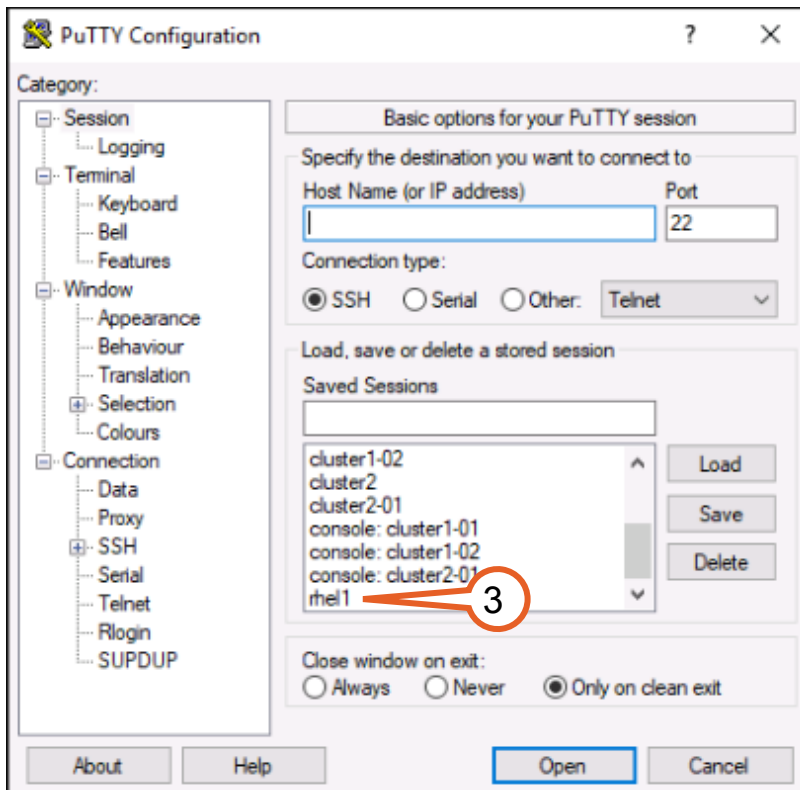


Figure 2-3:

A PuTTY session window for **rhel1** opens.

4. Log in using the following credentials:
 - Username: **root**
 - Password: **Netapp1!**
5. This lab activity targets the “largeFG” FlexGroup volume that was created and populated during a session of the FlexGroup Fundamentals lab. Examine the “largeFG” volume’s contents.

```
[root@rhel1 ~]# cd /svm1
[root@rhel1 svm1]# ls
balanceFG convertFV demoFG largeFG smallFG
[root@rhel1 svm1]# cd largeFG
[root@rhel1 largeFG]# ls
dir_0 dir_1 dir_2 dir_3 largedir_0 largedir_1
[root@rhel1 largeFG]# find . | wc -l
```

```

104023
[root@rhell largeFG]# ls dir_0
subdir_000  subdir_167  subdir_334  subdir_501  subdir_668  subdir_835
subdir_001  subdir_168  subdir_335  subdir_502  subdir_669  subdir_836
subdir_002  subdir_169  subdir_336  subdir_503  subdir_670  subdir_837
.
# Omitted for brevity
.
subdir_163  subdir_330  subdir_497  subdir_664  subdir_831  subdir_998
subdir_164  subdir_331  subdir_498  subdir_665  subdir_832  subdir_999
subdir_165  subdir_332  subdir_499  subdir_666  subdir_833
subdir_166  subdir_333  subdir_500  subdir_667  subdir_834
[root@rhell largeFG]# ls dir_0/subdir_000
file_00.txt  file_05.txt  file_10.txt  file_15.txt  file_20.txt
file_01.txt  file_06.txt  file_11.txt  file_16.txt  file_21.txt
file_02.txt  file_07.txt  file_12.txt  file_17.txt  file_22.txt
file_03.txt  file_08.txt  file_13.txt  file_18.txt  file_23.txt
file_04.txt  file_09.txt  file_14.txt  file_19.txt  file_24.txt
[root@rhell largeFG]# ls largedir_0
largefile_0.dat  largefile_2.dat  largefile_4.dat  largefile_6.dat
largefile_1.dat  largefile_3.dat  largefile_5.dat  largefile_7.dat
[root@rhell largeFG]#

```

Specifically, the “largeFG” FlexGroup volume contains six directories:

- The four dir_* directories each contain 1000 subdirectories, with each subdirectory containing 25 small files filled with random text, for a total of 100,000 small files spread across 4004 directories.
- The two largedir_* directories each contain eight 1 GB data files filled with random binary data, for a total of 16 large files.

6. On the task bar of Jumphost, right-click **PuTTY** and select **PuTTY** from the context menu.

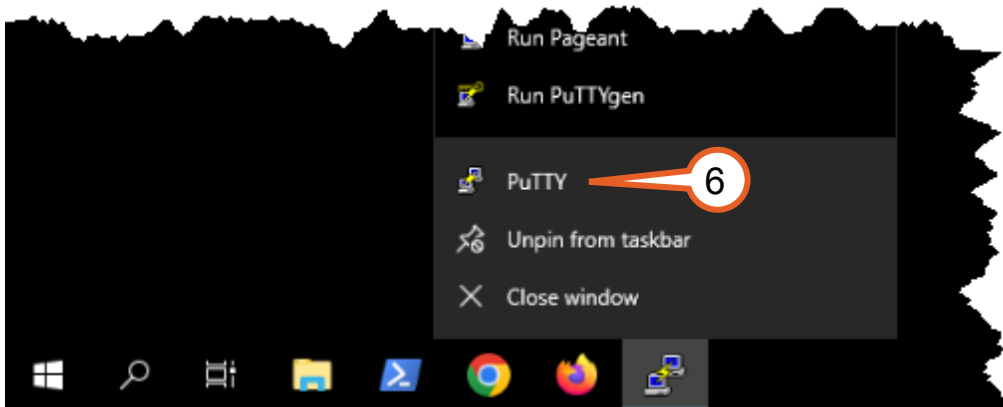


Figure 2-4:

- In the PuTTY Configuration window, double-click **cluster1** in the “Saved Sessions” list.

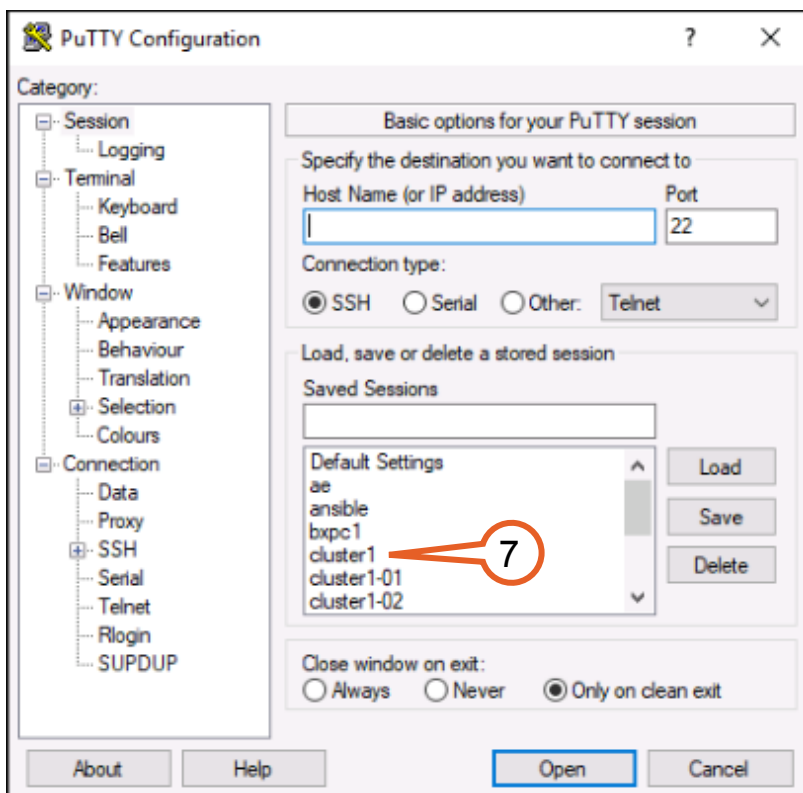


Figure 2-5:

- Log in using the following credentials:
 - Username: **admin**
 - Password: **Netapp1!**
- The largeFG FlexGroup volume has a configured capacity of approximately 600 GB and is comprised of four constituent volumes.

```
cluster1::> volume show -vserver svml
Vserver  Volume      Aggregate      State      Type      Size      Available Used%
-----
svml     balanceFG   -              online     RW        20GB      10.38GB  45%
svml     convertFV   aggr1_cluster1_01 online     RW        20GB      13.11GB  30%
svml     demoFG      -              online     RW        200GB     186.8GB  1%
svml     largeFG     -              online     RW        631.6GB   583.6GB  2%
svml     smallFG     -              online     RW        10GB      9.37GB   1%
svml     svml_root   aggr1_cluster1_01 online     RW        20MB      17.54MB  7%
6 entries were displayed.

cluster1::> volume show -vserver svml -volume largeFG* -is-constituent true
Vserver  Volume      Aggregate      State      Type      Size      Available Used%
-----
svml     largeFG__0001 aggr1_cluster1_01 online     RW        157.9GB   141.9GB   5%
svml     largeFG__0002 aggr1_cluster1_01 online     RW        157.9GB   149.9GB   0%
svml     largeFG__0003 aggr1_cluster1_02 online     RW        157.9GB   149.9GB   0%
svml     largeFG__0004 aggr1_cluster1_02 online     RW        157.9GB   141.9GB   5%
4 entries were displayed.
```

```
cluster1::>
```

10. Examine the file count and volume capacity metrics for each of largeFG's member volumes.

```
cluster1::> set -priv diag

Warning: These diagnostic commands are for use by NetApp personnel only.
Do you want to continue? {y|n}: y

cluster1::> vol show -vserver svml -volume largeFG* -fields size,used,percent-used,files,files-used
vserver volume size used percent-used files files-used
-----
svml largeFG 631.6GB 16.37GB 2% 19660752 104412
svml largeFG_0001 157.9GB 8.11GB 5% 4915188 26119
svml largeFG_0002 157.9GB 79.21MB 0% 4915188 26135
svml largeFG_0003 157.9GB 76.33MB 0% 4915188 26053
svml largeFG_0004 157.9GB 8.11GB 5% 4915188 26105
5 entries were displayed.

cluster1::>
```

The “files-used” counts for the four constituent volumes are close to equal, meaning the largeFG FlexGroup is in balance. The constituent volume storage space “used” values do show a fairly significant difference, with two constituents at ~500 MB of space used while the other two are at ~8 GB. This is an artifact of the specific data set contained within the “largeFG” FlexGroup volume.

When ingesting files to a FlexGroup volume that is currently in balance, ONTAP prefers to place new files on the same constituent volume as their containing directory. In this example, the populated data set’s two largedir_* directories were placed on separate constituent volumes, and each directory’s set of eight 1 GB files were placed on the same constituent volume as their parent directory. In other words, two constituent volumes each received a largedir directory and its complete set of large binary files, while the two remaining constituent volumes did not. The many small text file directories and their files were placed in a similar manner, but since there are so many more of those directories, all of the constituent volumes received some of those directories.

11. ONTAP uses a sophisticated algorithm for determining where to place the next ingested file. Examine the metrics that ONTAP dynamically calculates in support of that algorithm.

```
cluster1::> node run * flexgroup show largeFG
2 entries were acted on.

Node: cluster1-01
Command time: Sat May 13 17:06:43 UTC 2023

FlexGroup 0x804FF6D1 (largeFG)
* next snapshot cleanup due in 6387 msec
* next refresh message due in 411 msec (last to member 0x804FF6D5)
* spinnp version negotiated as 14.11, capability 0x1FF7F
* Ref count is 4
* ShouldEnforceQuotas false
* IsAnyMemberInNvfailedState false
* reaction +0.0, workload +0.0, activity level 0, cv 0%, width 512MB
* Rebalancing configuration:
    max_moves=0, max_threshold=0, min_threshold=0
    min_file_size=0, max_runtime=0, exclude_snapshots=true
    start_time=0, stop_time=0
```

Idx	Member	L	Used	Avail	Urgc	Target	Probabilities	D-Ingest	Alloc	F-	
1	1032	L	2118610	5%	39313409	0%	22.72%	[100% 100% 77% 77%]	0+	0	0
0+	0	0									
2	1033	L	12087	0%	39313409	0%	27.27%	[100% 100% 72% 72%]	0+	0	0
0+	0	0									
3	1034	R	11349	0%	39313409	0%	27.27%	[100% 100% 72% 72%]	0+	0	0
0+	0	0									
4	1035	R	2117515	5%	39313409	0%	22.72%	[100% 100% 77% 77%]	0+	0	0
0+	0	0									

```
Node: cluster1-02
Command time: Sat May 13 17:06:43 UTC 2023
```

```

FlexGroup 0x804FF6D1 (largeFG)
* next snapshot cleanup due in 7144 msec
* next refresh message due in 147 msec (last to member 0x804FF6D3)
* spinnp version negotiated as 14.11, capability 0x1FF7F
* some members have uninitialized fgindex values
* Ref count is 4
* ShouldEnforceQuotas false
* IsAnyMemberInNvfailedState false
* reaction +0.0, workload +0.0, activity level 0, cv 0%, width 512MB
* Rebalancing configuration:
    max_moves=0, max_threshold=0, min_threshold=0
    min_file_size=0, max_runtime=0, exclude_snapshots=true
    start_time=0, stop_time=0

```

Idx	Member	L	Used	Avail	Urgc	Target	Probabilities	D-Ingest	Alloc	F-	
1	1032	R	2118610	5%	39313409	0%	22.72%	[100% 100% 77% 77%]	0+	0	0
0+	0	0									
2	1033	R	12087	0%	39313409	0%	27.27%	[100% 100% 72% 72%]	0+	0	0
0+	0	0									
3	1034	L	11349	0%	39313409	0%	27.27%	[100% 100% 72% 72%]	0+	0	0
0+	0	0									
4	1035	L	2117515	5%	39313409	0%	22.72%	[100% 100% 77% 77%]	0+	0	0
0+	0	0									

```

cluster1::*>

```

The two most important metrics to pay attention to here are “Target” and “Urgc”:

- “Target” is the target percentage for file placement on the given constituent volume under normal conditions. If you add up all of the target percentage values for a given FlexGroup volume’s constituents, the total should come to 100% (after accounting for display rounding errors). Ideally, every constituent volume will have the same target percentage value, but if one constituent volume gets more data or file count than the others then all of the values will adjust to favor placement on other constituent volumes.
- “Urgc” is the urgency level percentage, and represents how often the placement algorithm will override normal target placement decisions. If a constituent volume’s capacity consumption exceeds 90%, then ONTAP will increase the urgency level so that new files will avoid that constituent volume more often. As an example, if a constituent volume’s target percentage is 25% and its Urgc is 80%, ONTAP would normally try to place a file locally to that constituent 25% of the time. However, 80% of the time it would override that local placement decision as an emergency measure to keep the FlexGroup in balance, and so it would instead place the file to a remote constituent volume.

12. Set the ONTAP CLI privilege back to the standard level.

```

cluster1::*> set -priv admin
cluster1::>

```

Takeaway: ONTAP automatically distributes newly ingested files evenly across constituent volumes while maintaining a balance of constituent volume available capacity.

2.2 Exploring FlexGroup Autosize Behavior

The data sets for a business’s big data applications frequently evolve, often growing rapidly in unpredictable ways. A storage architect needs to implement a cost-effective big data storage solution nimble enough to keep up with these changing requirements.

When you create files on a FlexGroup volume, ONTAP maps a given file’s entire contents to just one of the FlexGroup’s constituent volumes. That file’s entire contents will continue to be hosted on that same single constituent volume during subsequent files updates, up until the file is deleted, or possibly until the ONTAP administrator initiates a FlexGroup volume rebalance operation (a feature discussed in a later section of this lab). An outgrowth of this behavior is that a constituent volume could potentially fill up due to organic growth in its existing files, despite the availability of free space in the FlexGroup’s other constituent volumes.

ONTAP includes several FlexGroup autosize features that reduce the possibility of an individual FlexGroup constituent volume filling up in this manner, and of an entire FlexGroup running out of space.

- Elastic sizing intervenes just before a file write on a constituent volume would otherwise issue an “out of space” error. Before issuing that error, ONTAP attempts to “borrow” a limited amount of free space from other constituent volumes so that the write can complete without needing to grow the FlexGroup.
- Proactive resizing encourages consistent FlexGroup performance and capacity by striving to maintain a relatively equal distribution of available capacity across all of a given FlexGroup’s constituent volumes. It monitors the capacity of FlexGroup constituent volumes and dynamically sizes them up and down (within the constraints of the FlexGroup’s existing overall capacity) in response to changes in their contained file space consumption.
- Volume autogrow allows ONTAP to dynamically change the size of a volume (FlexGroup or FlexVol) as those volumes cross certain configurable capacity thresholds. The feature is managed on a per-volume basis, and can be disabled, configured to only grow the volume, or configured to both grow and shrink the volume.



Tip: These autosize features are not a panacea. They provide some protection against failed writes or corrupted files stemming from surprise sudden growth scenarios, but they are not a substitute for conducting regular storage administration due diligence. NetApp recommends that you regularly monitor the available capacity of your constituent volumes and that you take steps to generally maintain their capacity below 80% full.

This lab activity demonstrates how a FlexGroup volume and its constituent volumes behave when growth scenarios bring these autosize features into play.

1. During this lab activity you will frequently switch between your PuTTY sessions for “cluster1” and “rhe1”. Arrange those PuTTY sessions on your display so that you can simultaneously view the full contents of both windows.
2. In your PuTTY session for “cluster1”, examine the smallFG FlexGroup volume that you will be using in this lab activity.

```
cluster1::> volume show -vserver svm1
Vserver  Volume      Aggregate      State      Type      Size      Available  Used%
-----
svm1     balanceFG   -              online     RW         20GB      10.32GB    45%
svm1     convertFV   aggr1_cluster1_01
          online     RW         20GB      13.11GB    30%
svm1     demoFG      -              online     RW         200GB     186.8GB    1%
svm1     largeFG     -              online     RW         631.6GB   583.6GB    2%
svm1     smallFG     -              online     RW         10GB      9.37GB     1%
svm1     svm1_root   aggr1_cluster1_01
          online     RW         20MB      17.51MB    7%

6 entries were displayed.

cluster1::> volume show -vserver svm1 -volume smallFG* -is-constituent true
Vserver  Volume      Aggregate      State      Type      Size      Available  Used%
-----
svm1     smallFG__0001
          aggr1_cluster1_01
          online     RW         2.50GB     2.34GB     1%
svm1     smallFG__0002
          aggr1_cluster1_02
          online     RW         2.50GB     2.34GB     1%
svm1     smallFG__0003
          aggr1_cluster1_01
          online     RW         2.50GB     2.34GB     1%
svm1     smallFG__0004
          aggr1_cluster1_02
          online     RW         2.50GB     2.34GB     1%

4 entries were displayed.

cluster1::>
```



Note: This FlexGroup volume deviates from NetApp best practices in the following ways.

- The “smallFG” FlexGroup volume is comprised of four constituent volumes that are much smaller (2.5 GB) than the recommended 100 GB minimum. Later in this lab activity you will populate large files to this FlexGroup volume to see how the constituent volumes behave as they get full. Using larger volumes would require you to populate much larger files, and endure correspondingly longer wait times during that population process.

- Snapshot are turned off on the “smallFG” FlexGroup volume. In the late stages of this demonstration you will delete a large file in order to observe the volume autogrow feature reduce the size of the FlexGroup volume. That observation is not immediately possible if snapshots are enabled and a snapshot initiated prior to the file delete action. To be clear, snapshots and volume autogrow, including volume shrink, co-exist just fine, it’s just that ONTAP will not releases the blocks for deleted files that are captured in snapshots until such time as the snapshot is deleted.
- Volume autosize is disabled.

3. Elevate your ONTAP CLI session privilege level to diag mode.

```
cluster1::> set -priv diag

Warning: These diagnostic commands are for use by NetApp personnel only.
Do you want to continue? {y|n}: y

cluster1::*>
```

4. Display more detailed constituent volume information for the smallFG FlexGroup.

```
cluster1::*> volume show -vserver svm1 -volume smallFG* -fields size,available,used,percent-
used,files,files-used
vserver volume size available used percent-used files files-used
-----
svm1 smallFG 10GB 9.37GB 129.8MB 1% 311276 387
svm1 smallFG_0001
2.50GB
2.34GB 32.45MB 1% 77819 96
svm1 smallFG_0002
2.50GB
2.34GB 32.45MB 1% 77819 97
svm1 smallFG_0003
2.50GB
2.34GB 32.44MB 1% 77819 97
svm1 smallFG_0004
2.50GB
2.34GB 32.45MB 1% 77819 97
5 entries were displayed.

cluster1::*>
```

The volume sizes, used space, and file counts for each of the constituent volumes are all essentially the same.

5. In your PuTTY session for “rhel1”, change directory into the local NFS mount point for the smallFG FlexGroup volume, list its contents, and verify that the volume is NFS-mounted.

```
[root@rhel1 svm1]# cd /svm1/smallFG
[root@rhel1 smallFG]# ls
[root@rhel1 smallFG]# df -h .
Filesystem Size Used Avail Use% Mounted on
svm1:/smallFG 9.5G 130M 9.4G 2% /svm1/smallFG
[root@rhel1 smallFG]#
```

The volume is empty, and the “svm1:/smallFG” filesystem path is typical for an NFS-mounted volume.

6. Create a 3 GB file and a 4 GB file on the smallFG FlexGroup volume.



Important: Enter the following entire command as a single line and immediately proceed to the next step, without waiting for the command to finish.

```
[root@rhel1 smallFG]# time dd if=/dev/urandom count=3000 bs=1M conv=fsync of=3G-file.dat; time dd
if=/dev/urandom count=4000 bs=1M conv=fsync of=4G-file.dat
3000+0 records in
3000+0 records out
3145728000 bytes (3.1 GB, 2.9 GiB) copied, 20.8749 s, 151 MB/s

real 0m20.919s
user 0m0.005s
sys 0m19.852s
4000+0 records in
4000+0 records out
4194304000 bytes (4.2 GB, 3.9 GiB) copied, 28.5 s, 147 MB/s
```

```
real    0m28.692s
user    0m0.011s
sys     0m25.912s
[root@rhell smallFG]#
```

7. In your PuTTY session for “cluster1”, periodically (every 30 seconds or so) re-display the constituent volume information for the smallFG FlexGroup while the dd commands from the previous step run, and continue doing so after they finish until the output stabilizes (typically takes a 2-3 minutes).



Tip: This is the same “volume show” command used in step 4. You can use the up-arrow key to recall that previous command for re-execution here.

```
cluster1::*> # This time point corresponds to shortly after the dd commands started.

cluster1::*> volume show -vserver svml -volume smallFG* -fields size,available,used,percent-
used,files,files-used
vserver volume size available used percent-used files files-used
-----
svml smallFG 10GB 7.99GB 1.51GB 15% 311276 391
svml smallFG_0001
2.50GB
2.34GB 32.47MB
1% 77819 97
svml smallFG_0002
2.50GB
2.34GB 32.45MB
1% 77819 97
svml smallFG_0003
2.50GB
839.5MB 1.55GB 65% 77819 100
svml smallFG_0004
2.50GB
2.34GB 32.45MB
1% 77819 97
5 entries were displayed.

cluster1::*> # This time point corresponds to about one minute, the time the dd commands completed...

cluster1::*> volume show -vserver svml -volume smallFG* -fields size,available,used,percent-
used,files,files-used
vserver volume size available used percent-used files files-used
-----
svml smallFG 10.20GB 2.70GB 6.99GB 72% 383402 396
svml smallFG_0001
1.40GB 1.29GB 32.54MB
2% 77819 98
svml smallFG_0002
1.29GB 1.19GB 32.71MB
2% 77819 97
svml smallFG_0003
3.14GB 13.38MB 2.97GB 99% 97862 100
svml smallFG_0004
4.17GB 10.25MB 3.95GB 99% 129902 101
5 entries were displayed.

cluster1::*> # This time point corresponds to about two minutes after the dd commands completed...

cluster1::*> volume show -vserver svml -volume smallFG* -fields size,available,used,percent-
used,files,files-used
vserver volume size available used percent-used files files-used
-----
svml smallFG 10GB 2.51GB 6.99GB 73% 440007 396
svml smallFG_0001
714.6MB
646.3MB 32.56MB
4% 77819 98
svml smallFG_0002
711.8MB
643.5MB 32.73MB
4% 77819 97
svml smallFG_0003
3.78GB
635.1MB 2.97GB 82% 117759 100
svml smallFG_0004
4.82GB
643.4MB 3.95GB 86% 166610 101
5 entries were displayed.
```

```
cluster1::*>
```

As you review the output of the volume show commands, you should note the following developments:

- The 3 GB and 4 GB files are placed on different constituent volumes, detectable in this example by the reported used space consumption of the constituent volumes. The specific constituents used in your lab may differ from those shown in this example.
- ONTAP grows the size of the two constituent volumes, and reduces the size of the other constituent volumes to compensate.
- Although the available space values reported across the constituent volumes diverge by quite a bit immediately after the dd commands complete, they later converge to roughly equivalent values again (in this specific lab scenario, typically within one to two minutes after the dd commands complete) as ONTAP works to proactively balance the free space across the constituent volumes.
- The overall size of the FlexGroup volume remains at 10 GB.



Note: The artificially small constituent volumes used in this demonstration can produce rounding errors that generate small inconsistencies in the output of the “show volume” command. For example, the second “volume show” command in this example erroneously indicates that “smallFG” has grown to 10.20 GB in size, but later commands correctly indicate that it is still 10 GB.

8. In your PuTTY session for “rhel1”, now create a 5 GB file, which will exceed the remaining available storage capacity on the FlexGroup volume and therefore prematurely fail with a “no space left on device” error. List the sizes of the file on “smallFG” as confirmation of the error.

```
[root@rhel1 smallFG]# time dd if=/dev/urandom count=5000 bs=1M conv=fsync of=5G-file.dat
dd: error writing '5G-file.dat': No space left on device
dd: closing output file '5G-file.dat': No space left on device

real    0m25.253s
user    0m0.005s
sys     0m19.323s
[root@rhel1 smallFG]# ls -l
total 9810992
-rw-r--r--. 1 root root 3145728000 May 15 17:33 3G-file.dat
-rw-r--r--. 1 root root 4194304000 May 15 17:34 4G-file.dat
-rw-r--r--. 1 root root 2735603712 May 15 17:41 5G-file.dat
[root@rhel1 smallFG]#
```

The size of the 5G-file.dat file is much smaller than the anticipated 5 GB, further confirming the error.

9. In your PuTTY session for “cluster1”, once again display the constituent volume information for the smallFG FlexGroup.

```
cluster1::*> volume show -vserver svml -volume smallFG* -fields size,available,used,percent-
used,files,files-used
vserver volume  size available used  percent-used  files  files-used
-----
svml      smallFG 10GB 39.62MB  9.46GB 99%          444928 399
svml      smallFG_0001
          2.66GB
          12KB  2.52GB 99%          82740 102
svml      smallFG_0002
          30MB 20.63MB 27%          77819 97
svml      smallFG_0003
          3.14GB
          9.49MB 2.97GB 99%          117759 100
svml      smallFG_0004
          4.17GB
          9.49MB 3.95GB 99%          166610 100
5 entries were displayed.

cluster1::*>
```

The output shows that the volume is essentially full (it has less than 40 MB of free space). Observe that the “smallFG” FlexGroup volume’s overall size is still 10 GB.

10. Display the smallFG volume’s current autosize settings.

```
cluster1::*> volume show -vserver svml -volume smallFG -fields autosize-mode,max-autosize,autosize-
grow-threshold-percent
```

```

vserver volume max-autosize autosize-grow-threshold-percent autosize-mode
-----
svml      smallFG 12GB          85%                      off

cluster1::*>

```

Autosize is currently turned off for this volume. If it were turned on, autosize would only start trying to grow this volume once it exceeded 85% of overall capacity, but only so long as the volume's total size remains below the shown limit of 12 GB.

11. Turn on autosize combined grow and shrink mode for the smallFG FlexGroup volume, and set its maximum growth size to 20 GB.

```

cluster1::*> volume modify -vserver svml -volume smallFG -autosize-mode grow_shrink -max-autosize
20g
[Job 202] Job succeeded: volume modify succeeded

cluster1::*>

```

12. The smallFG FlexGroup will start automatically growing to alleviate the lack of available space. Periodically (every 30 seconds or so) re-query for the constituent volume details for the “smallFG” FlexGroup volume until the volume size stabilizes at about 11.85 GB, and for the constituent volumes to stabilize at roughly equal amounts of available space. In this lab that typically takes around two minutes.

```

cluster1::*> # This time point corresponds to roughly two minutes after enabling autosize-mode...

cluster1::*> volume show -vserver svml -volume smallFG* -fields size,available,used,percent-
used,files,files-used
vserver volume size available used percent-used files files-used
-----
svml      smallFG 11.85GB 1.78GB 9.49GB 84% 460502 396
svml      smallFG_0001 3.13GB 456.2MB 2.52GB 85% 98314 99
svml      smallFG_0002 497.4MB 440.1MB 32.42MB 6% 77819 97
svml      smallFG_0003 3.61GB 466.3MB 2.97GB 86% 117759 100
svml      smallFG_0004 4.63GB 455.6MB 3.95GB 89% 166610 100
5 entries were displayed.

cluster1::*>

```

13. In your PuTTY session for “rhel1”, re-try creating the 5 GB file. If you get a “no space left on device” error in the command output, see the note found after the code block..

```

[root@rhel1 smallFG]# time dd if=/dev/urandom count=5000 bs=1M conv=fsync of=5G-file.dat
dd: fsync failed for '5G-file.dat': No space left on device
5000+0 records in
5000+0 records out
5242880000 bytes (5.2 GB, 4.9 GiB) copied, 83.6748 s, 62.7 MB/s

real    1m23.845s
user    0m0.022s
sys     0m34.333s
total 12336396
[root@rhel1 smallFG]# ls -l
-rw-r--r--. 1 root root 3145728000 May 13 17:20 3G-file.dat
-rw-r--r--. 1 root root 4194304000 May 13 17:21 4G-file.dat
-rw-r--r--. 1 root root 5242880000 May 13 17:36 5G-file.dat
[root@rhel1 smallFG]

```



Note: The dd command in this step produces a “dd: fsync failed for '5G-file.dat': No space left on device” error message that you can safely ignore. This specific dd command creates a file containing 5000 1MB blocks, and the command output clearly shows that 5000 blocks were read in and 5000 blocks were written out. The final size of the 5G-file.dat file on the volume, 5242880000 bytes, which equals 5000 * 1024 * 1024, further confirms that all of the expected data was successfully written out to the file.



Tip: You may have noticed that the dd command for generating this 5 GB file took over 80 seconds to complete, while the dd command for the 4 GB file you created in step 6 took significantly less time, only

about 30 seconds. While enabling autosize provides an effective insurance policy against a FlexGroup or FlexVol volume filling up unexpectedly, the affected volume will experience a performance impact while autosize is dynamically growing the already full volume. That's an acceptable compromise when a sudden spike in space consumption would otherwise result in highly disruptive "out-of-space" write errors, but probably not something you want to live with for handling every day storage growth scenarios. A better solution for that is to proactively resize volumes well ahead of anticipated growth, as there is no performance impact from resizing a volume in that situation. As a general rule, once a volume reaches about 80% full you should start to think about adding capacity.

14. In your PuTTY session for "cluster1", again periodically display the constituent information for the smallFG FlexGroup, until each constituent volume's reported available space stabilizes at roughly equivalent values. In this lab that typically occurs within a couple of minutes after the dd command completes.

```
cluster1::*> # This time point corresponds to about 2 minutes after the dd command completed.

cluster1::*> volume show -vserver svml -volume smallFG* -fields size,available,used,percent-
used,files,files-used
vserver volume size available used percent-used files files-used
-----
svml smallFG 15.06GB 2.41GB 11.90GB 83% 553176 396
svml smallFG_0001 5.84GB 623.9MB 4.94GB 89% 190709 99
svml smallFG_0002 685.8MB 619.1MB 32.44MB 4% 77819 97
svml smallFG_0003 3.77GB 618.6MB 2.97GB 83% 118038 100
svml smallFG_0004 4.79GB 607.5MB 3.95GB 86% 166610 100
5 entries were displayed.

cluster1::*>
```

The "smallFG" FlexGroup volume's total size has increased to about 15 GB.

15. In your PuTTY session for "rhel1", delete the 5G file.

```
[root@rhel1 smallFG]# pwd
/svml/smallFG
[root@rhel1 smallFG]# ls
3G-file.dat 4G-file.dat 5G-file.dat
[root@rhel1 smallFG]# rm 5G-file.dat
rm: remove regular file '5G-file.dat'? y
[root@rhel1 smallFG]#t
```

16. In your PuTTY session for "cluster1", once again periodically display the constituent volume information for the smallFG FlexGroup volume. When ONTAP finishes releasing the underlying blocks from the deleted file, the FlexGroup begins both shrinking the FlexGroup volume and re-balancing the available space across the constituent volumes. In this lab activity it typically takes two to three minutes for autosize to stabilize the volume size.

```
cluster1::*> # This time point corresponds to approximately 20 seconds after the file was deleted...

cluster1::*> volume show -vserver svml -volume smallFG* -fields size,available,used,percent-
used,files,files-used
vserver volume size available used percent-used files files-used
-----
svml smallFG 15.06GB 7.12GB 7.19GB 50% 553176 395
svml smallFG_0001 4.65GB 4.19GB 237.9MB 5% 190709 98
svml smallFG_0002 685.8MB 619.1MB 32.45MB 4% 77819 97
svml smallFG_0003 3.77GB 618.6MB 2.97GB 83% 118038 100
svml smallFG_0004 4.97GB 788.0MB 3.95GB 83% 166610 100
5 entries were displayed.

cluster1::*> # This time point corresponds to approximately 40 seconds after the file was deleted...

cluster1::*> volume show -vserver svml -volume smallFG* -fields size,available,used,percent-
used,files,files-used
```

```

vserver volume size available used percent-used files files-used
-----
svm1 smallFG 15.06GB 7.12GB 7.19GB 50% 553176 395
svm1 smallFG_0001 4.65GB 4.19GB 237.9MB 5% 190709 98
svm1 smallFG_0002 685.8MB 619.1MB 32.45MB 4% 77819 97
svm1 smallFG_0003 3.77GB 618.6MB 2.97GB 83% 118038 100
svm1 smallFG_0004 4.97GB 788.0MB 3.95GB 83% 166610 100
5 entries were displayed.

cluster1::*> # This time point corresponds to approximately 3 minutes after the file was deleted...

cluster1::*> volume show -vserver svm1 -volume smallFG* -fields size,available,used,percent-
used,files,files-used
vserver volume size available used percent-used files files-used
-----
svm1 smallFG 11.42GB 3.86GB 6.99GB 64% 564109 395
svm1 smallFG_0001 1.06GB 1000MB 33.97MB 3% 190709 98
svm1 smallFG_0002 1.05GB 986.3MB 32.47MB 3% 77819 97
svm1 smallFG_0003 4.14GB 985.1MB 2.97GB 75% 128971 100
svm1 smallFG_0004 5.17GB 977.3MB 3.95GB 80% 166610 100
5 entries were displayed.

cluster1::*>

```

In your lab session “smallFG” might stabilize at a higher total size than shown in this example, but it will still be significantly less than the previous size of 15 GB. Most importantly, the available space for each of the constituent volumes in your lab will have reached an approximately equal value to each other.

If you were to delete one or both of the other remaining files from the volume, further “volume show” commands would report the “smallFG” FlexGroup volume shrinking even further, smaller than even smallFG’s original size of 10 GB.

17. Take your ONTAP CLI session out of diag privilege mode by switching to the “admin” privilege level, the ONTAP CLI’s default privilege level.

```

cluster1::*> set -priv admin

cluster1::>

```

Takeaway: ONTAP automatically resizes FlexGroup Volumes and their constituents, enabling each constituent to have equal access to available capacity. Automatic resizing also supports cost effective storage allocation by reducing over-provisioning of excessive capacity to existing constituents while eliminating the time and labor otherwise required for the storage administrators to implement those changes.

2.3 Add New Constituent Volumes to an Existing FlexGroup

You can add new constituent volumes to an existing FlexGroup volume without any adverse impact to the FlexGroup volume’s availability or performance. If that FlexGroup volume is in a pre-existing SnapMirror relationship, ONTAP will also automatically add the new constituent volume to the SnapMirror destination during the next update cycle, without any need to re-create the relationship or re-transfer all of the FlexGroup volume’s data.

In this lab activity a storage administrator wants to understand the effects of adding a new constituent volume to an existing FlexGroup volume. You add four additional constituent volumes to the largeFG FlexGroup volume and observe how the volume’s pre-existing data protection relationship automatically adds the new constituent volumes to the destination volume during the next SnapMirror update. You also examine the impact that adding constituent volumes has on pre-existing volume snapshots.

1. In your PuTTY session for “cluster1”, elevate your privilege level to diagnostics mode, and then list the constituent volumes for the largeFG FlexGroup volume.

```
cluster1::> set -priv diag

Warning: These diagnostic commands are for use by NetApp personnel only.
Do you want to continue? {y|n}: y

cluster1::*> vol show -vserver svml -volume largeFG*
Vserver   Volume           Aggregate          State    Type    Size    Available  Used%
-----
svml      largeFG           -                  online   RW      631.6GB  581.9GB    3%
svml      largeFG__0001     aggr1_cluster1_01 online   RW      157.9GB  141.5GB    5%
svml      largeFG__0002     aggr1_cluster1_01 online   RW      157.9GB  149.5GB    0%
svml      largeFG__0003     aggr1_cluster1_02 online   RW      157.9GB  149.5GB    0%
svml      largeFG__0004     aggr1_cluster1_02 online   RW      157.9GB  141.5GB    5%

5 entries were displayed.

cluster1::*>
```

“LargeFG” has four constituent volumes.

2. Open a PuTTY a session to “cluster2”. On the task bar of Jumphost, right-click the **PuTTY icon** and select **PuTTY** from the context menu.

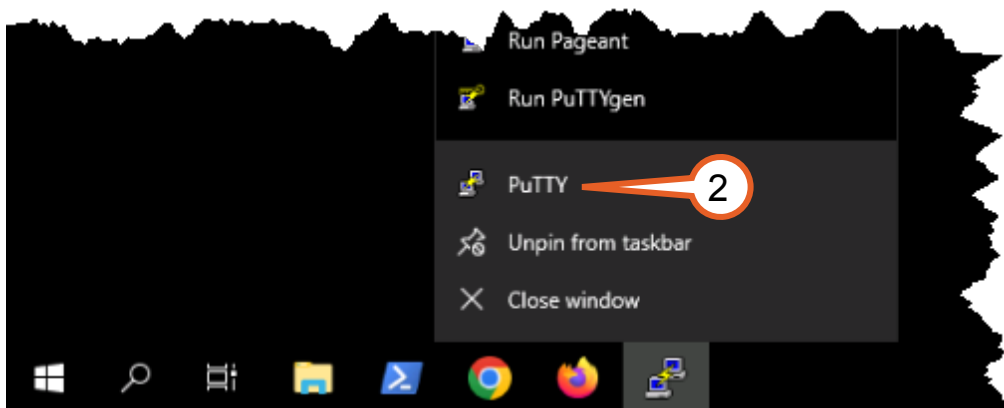


Figure 2-6:

The “PuTTY Configuration” dialog opens.

- Double-click **cluster2** in the “Saved Sessions” list.

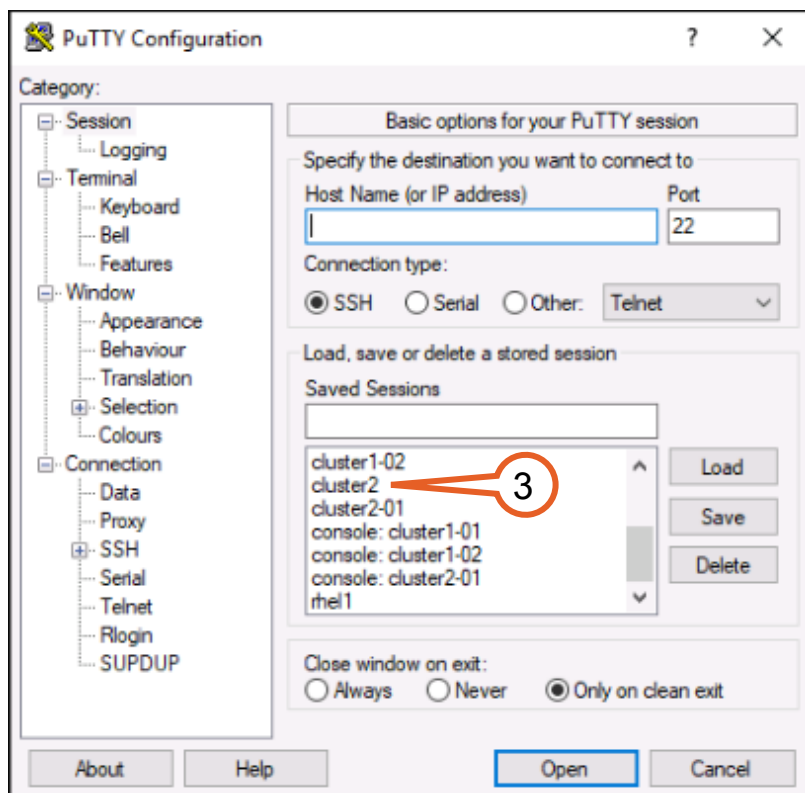


Figure 2-7:

The “PuTTY Configuration” dialog closes, and a PuTTY session window opens for cluster2.

- Log in using the following credentials:

- Username: **admin**
- Password: **Netapp1!**

On your display, arrange the PuTTY windows for cluster1 and cluster2 so you can simultaneously view the full contents of both windows. Maintaining simultaneous visibility to the rhel1 window is not necessary.

- In your PuTTY session for “cluster2”, verify that there is already an existing SnapMirror relationship from the largeFG volume on svm1 (on cluster1) to the vol_largeFG_dest volume on svm1-dest (on cluster2).

```
cluster2::> snapmirror show
```

Source Path	Type	Destination Path	Mirror State	Relationship Status	Total Progress	Progress Healthy	Last Updated
svm1:largeFG	XDP	svm1-dst:vol_largeFG_dest	Snapmirrored	Idle	-	true	-

```
cluster2::>
```

- Elevate your privilege level to diag mode and list the constituent volumes for the vol_largeFG_dest FlexGroup volume.

```
cluster2::> set -priv diag
```

Warning: These diagnostic commands are for use by NetApp personnel only.
Do you want to continue? {y|n}: **y**

```
cluster2::*> volume show -vserver svm1-dst -volume vol_largeFG_dest*
```

Vserver	Volume	Aggregate	State	Type	Size	Available	Used%
svm1-dst	vol_largeFG_dest						

```

-               online      DP      631.6GB    583.6GB    2%
svml1-dst  vol_largeFG_dest__0001
            aggr2_cluster2_01
            online      DP      157.9GB    141.9GB    5%
svml1-dst  vol_largeFG_dest__0002
            aggr2_cluster2_01
            online      DP      157.9GB    149.9GB    0%
svml1-dst  vol_largeFG_dest__0003
            aggr2_cluster2_01
            online      DP      157.9GB    149.9GB    0%
svml1-dst  vol_largeFG_dest__0004
            aggr2_cluster2_01
            online      DP      157.9GB    141.9GB    5%
5 entries were displayed.
cluster2::*>

```

7. In your PuTTY session for “cluster1”, add four new constituent volumes to the largeFG FlexGroup volume.

```

cluster1::*> vol expand -vserver svml -volume largeFG -aggr-list aggr1_cluster1_01,aggr1_cluster1_02
               -aggr-list-multiplier 2

Warning: The following number of constituents of size 157.9GB will be added to
         FlexGroup volume "largeFG": 4. Expanding the FlexGroup volume will
         cause the state of all Snapshot copies to be set to "partial". Partial
         Snapshot copies cannot be restored.
Do you want to continue? {y|n}: y
[Job 234] Job succeeded: Successful

cluster1::*>

```

8. List the constituent volumes for the largeFG FlexGroup volume.

```

cluster1::*> vol show -vserver svml -volume largeFG*
Vserver  Volume      Aggregate      State      Type      Size      Available  Used%
-----
svml1    largeFG     -               online     RW        1.23TB    1.15TB    1%
svml1    largeFG__0001
            aggr1_cluster1_01
            online     RW        157.9GB   141.5GB   5%
svml1    largeFG__0002
            aggr1_cluster1_01
            online     RW        157.9GB   149.5GB   0%
svml1    largeFG__0003
            aggr1_cluster1_02
            online     RW        157.9GB   149.5GB   0%
svml1    largeFG__0004
            aggr1_cluster1_02
            online     RW        157.9GB   141.5GB   5%
svml1    largeFG__0005
            aggr1_cluster1_01
            online     RW        157.9GB   149.5GB   0%
svml1    largeFG__0006
            aggr1_cluster1_02
            online     RW        157.9GB   149.5GB   0%
svml1    largeFG__0007
            aggr1_cluster1_01
            online     RW        157.9GB   149.5GB   0%
svml1    largeFG__0008
            aggr1_cluster1_02
            online     RW        157.9GB   149.5GB   0%
9 entries were displayed.
cluster1::*>

```

9. In your PuTTY session for “cluster2”, initiate an update of the SnapMirror relationship.

```

cluster2::*> snapmirror update -destination-path svml1-dst:vol_largeFG_dest
Operation is queued: snapmirror update of destination "svml1-dst:vol_largeFG_dest".

cluster2::*>

```

10. Periodically query the SnapMirror relationship state and wait for the update to finish (i.e., for the relationship status to change to “Idle”). In this lab activity, that update usually requires about one to two minutes to complete.

```
cluster2::*> snapmirror show
```

Progress

```

Source      Destination Mirror Relationship Total      Last
Path        Type    Path      State   Status   Progress Healthy Updated
-----
svm1:largeFG XDP    svm1-dst:vol_largeFG_dest
                               Snapmirrored
                               Transferring -         true    -

cluster2::*> snapmirror show

Source      Destination Mirror Relationship Total      Progress
Path        Type    Path      State   Status   Progress Healthy Updated
-----
svm1:largeFG XDP    svm1-dst:vol_largeFG_dest
                               Snapmirrored
                               Idle         -         false   -

cluster2::*>

```

11. List the constituent volumes for the destination FlexGroup volume, “vol_largeFG_dest”.

```

cluster2::*> vol show -vserver svm1-dst -volume vol_largeFG*
Vserver    Volume      Aggregate      State   Type      Size      Available Used%
-----
svm1-dst   vol_largeFG_dest
                               -         online    DP        1.23TB    1.16TB    1%
svm1-dst   vol_largeFG_dest_0001
                               aggr2_cluster2_01
                               online    DP        157.9GB   141.9GB   5%
svm1-dst   vol_largeFG_dest_0002
                               aggr2_cluster2_01
                               online    DP        157.9GB   149.9GB   0%
svm1-dst   vol_largeFG_dest_0003
                               aggr2_cluster2_01
                               online    DP        157.9GB   149.9GB   0%
svm1-dst   vol_largeFG_dest_0004
                               aggr2_cluster2_01
                               online    DP        157.9GB   141.9GB   5%
svm1-dst   vol_largeFG_dest_0005
                               aggr1_cluster2_01
                               online    DP        157.9GB   149.9GB   0%
svm1-dst   vol_largeFG_dest_0006
                               aggr1_cluster2_01
                               online    DP        157.9GB   149.9GB   0%
svm1-dst   vol_largeFG_dest_0007
                               aggr1_cluster2_01
                               online    DP        157.9GB   149.9GB   0%
svm1-dst   vol_largeFG_dest_0008
                               aggr1_cluster2_01
                               online    DP        157.9GB   149.9GB   0%

9 entries were displayed.

cluster2::*>

```

The new constituent volumes are now present on the data protection relationship destination volume as well.

12. In your PuTTY session for “cluster1” check the status of the snapshots on the largeFG FlexGroup volume.

```

cluster1::*> snapshot show -vserver svm1 -volume largeFG -fields size,blocks,usedblocks,state
vserver volume snapshot size blocks usedblocks state
-----
svm1     largeFG base      16.12GB 1% 50% partial
svm1     largeFG daily.2023-04-28_0010
                               16.12GB 1% 50% partial
svm1     largeFG daily.2023-04-29_0010
                               16.12GB 1% 50% partial
svm1     largeFG hourly.2023-04-29_1605
                               16.13GB 1% 50% partial
svm1     largeFG hourly.2023-04-29_1705
                               16.13GB 1% 50% partial
svm1     largeFG hourly.2023-05-13_0505
                               16.12GB 1% 50% partial
svm1     largeFG hourly.2023-05-13_1705
                               16.12GB 1% 50% partial
svm1     largeFG hourly.2023-05-13_1805
                               16.12GB 1% 50% partial
svm1     largeFG hourly.2023-05-13_1905
                               16.12GB 1% 50% valid

```

```
svml      largeFG snapmirror.e6a71223-aef8-11ed-8c75-005056b0ebc2_2162646627.2023-05-13_190726
          16.13GB 1%      50%      valid
10 entries were displayed.

cluster1::*>
```

Snapshots that pre-date your addition of the new constituent volumes to largeFG are marked as “partial”. Since these snapshots are unaware of those new constituent volumes, a wholesale restore of the whole FlexGroup volume from one of these snapshots would invalidate those new constituent volumes. However, ONTAP still allows you to use a storage client host to retrieve files from “partial” snapshots.

13. In your PuTTY session for “rhel1”, navigate to the largeFG volume’s dir_0/subdir_000 directory and delete the file_00.txt file.

```
[root@rhel1 smallFG]# cd /svml/largeFG/dir_0/subdir_000
[root@rhel1 subdir_000]# ls
file_00.txt  file_05.txt  file_10.txt  file_15.txt  file_20.txt
file_01.txt  file_06.txt  file_11.txt  file_16.txt  file_21.txt
file_02.txt  file_07.txt  file_12.txt  file_17.txt  file_22.txt
file_03.txt  file_08.txt  file_13.txt  file_18.txt  file_23.txt
file_04.txt  file_09.txt  file_14.txt  file_19.txt  file_24.txt
[root@rhel1 subdir_000]# ls -l file_00.txt
-rw-r--r--. 1 root root 33 Apr 27 18:07 file_00.txt
[root@rhel1 subdir_000]# rm file_00.txt
rm: remove regular file 'file_00.txt'? y
[root@rhel1 subdir_000]# ls
file_01.txt  file_05.txt  file_09.txt  file_13.txt  file_17.txt  file_21.txt
file_02.txt  file_06.txt  file_10.txt  file_14.txt  file_18.txt  file_22.txt
file_03.txt  file_07.txt  file_11.txt  file_15.txt  file_19.txt  file_23.txt
file_04.txt  file_08.txt  file_12.txt  file_16.txt  file_20.txt  file_24.txt
[root@rhel1 subdir_000]#
```

14. Use “rhel1” to restore the file from the volume’s “base” snapshot.

```
[root@rhel1 subdir_000]# ls .snapshot
base
daily.2023-04-28_0010
daily.2023-04-29_0010
hourly.2023-04-29_1605
hourly.2023-04-29_1705
hourly.2023-05-13_0505
hourly.2023-05-13_1705
hourly.2023-05-13_1805
hourly.2023-05-13_1905
snapmirror.e6a71223-aef8-11ed-8c75-005056b0ebc2_2162646627.2023-05-13_190726
[root@rhel1 subdir_000]# ls .snapshot/base
file_00.txt  file_05.txt  file_10.txt  file_15.txt  file_20.txt
file_01.txt  file_06.txt  file_11.txt  file_16.txt  file_21.txt
file_02.txt  file_07.txt  file_12.txt  file_17.txt  file_22.txt
file_03.txt  file_08.txt  file_13.txt  file_18.txt  file_23.txt
file_04.txt  file_09.txt  file_14.txt  file_19.txt  file_24.txt
[root@rhel1 subdir_000]# ls -l .snapshot/base/file_00.txt
-rw-r--r--. 1 root root 33 Apr 27 18:07 .snapshot/base/file_00.txt
[root@rhel1 subdir_000]# cp -p .snapshot/base/file_00.txt .
[root@rhel1 subdir_000]# ls -l file_00.txt
-rw-r--r--. 1 root root 33 Apr 27 18:07 file_00.txt
[root@rhel1 subdir_000]#
```

Remember that a file will reside on the same constituent volume throughout its lifespan (unless a storage administrator initiates a FlexGroup rebalance, discussed in an upcoming lab activity). When you deleted the file during the previous step you ended that file’s lifespan. Using rhel1 to copy the file from the FlexGroup volume’s .snapshot directory to its original location is actually a new file ingestion, meaning the newly restored file may now reside on one of the FlexGroup’s other constituent volumes rather than on the original constituent volume. As you saw during the earlier “File Layout and Balance” lab activity, when a FlexGroup volume is in balance, ONTAP favors placing a file on the same constituent volume as its parent directory, but that is not a guaranteed outcome.

Takeaway: ONTAP makes it fast and easy for storage administrators to add new constituent volumes to an existing FlexGroup volume without the need to modify existing data protection relationships. When storage users subsequently create new files on the FlexGroup volume, those new files automatically favor placement on the new constituent volumes.

2.4 Automatically Rebalance a FlexGroup Volume

Automatic FlexGroup volume rebalancing is a new feature introduced in ONTAP 9.12.1 to address problematic capacity imbalances that may arise within a FlexGroup volume's constituent volumes. There are several aspects to FlexGroup rebalancing that you need to understand in order to use this capability effectively.

First, the presence of a FlexGroup Volume capacity imbalance is not a problem in and of itself. Some level of imbalance in a FlexGroup is perfectly normal, and even bigger imbalances are often transient, tending to resolve themselves over time as storage clients delete files from and add files to the FlexGroup volume. (See [FlexGroup File Layout and Balance](#)). That said, some specific scenarios exist where it may make sense to initiate a rebalance operation on an imbalanced FlexGroup, including:

- As a follow-on to converting a FlexVol volume to a FlexGroup volume.
- After adding new constituent volumes to a pre-existing FlexGroup volume.
- If a FlexGroup volume's workload is experiencing significant performance bottlenecks as a result of an imbalance. This might occur if a majority of your workload's hot files have landed on the same constituent volume and those volumes have a long lifespan, meaning they have not been deleted and replaced (which would result in a new file ingestion).

Second, performing a rebalance on a FlexGroup volume involves a number of trade-offs, some of which have permanent consequences for the target FlexGroup volume. Those trade-offs include:

- You must enable multipart inode granular data functionality on the candidate FlexGroup volume. Once enabled, you will be unable to revert the hosting cluster to a previous ONTAP version unless you either revert the FlexGroup volume to a snapshot that pre-dates enabling granular data or you delete the FlexGroup volume.

Multipart inode granular data functionality, which is disabled by default for a FlexGroup volume, converts the FlexGroup volume to support multipart inodes. In a FlexGroup volume where granular data is disabled, a file's metadata and contents are stored together within the same [inode](#), all hosted on the same constituent volume. When granular data is enabled, a rebalance operation can convert an eligible file's inode to a multipart parent inode, redeploying the file's contents to one or more children inodes that it places on different constituent volumes from the parent inode. The parent inode continues to reside on the original constituent and hosts the file's metadata, retaining the original inode's filehandle, making rebalance operations non-disruptive to storage clients.

Importantly, new file ingestions to a FlexGroup volume that has granular data enabled produce traditional inodes, not multipart inodes.

- You manually start rebalance operations. A given FlexGroup volume only supports a single running rebalance operation at a time, but you can run rebalance operations for different FlexGroup volumes in parallel. A rebalance operation runs for a configurable period of time (6 hours by default) and self-terminates when that time window ends. When ONTAP is running a rebalance operation, it automatically selects the files to move, subject to file size and snapshot eligibility criteria supplied by the storage administrator, as well as the constituent volumes to move them to. A rebalance operation uses a scanning process to identify candidate files for rebalancing, and that scanning process is also responsible for relocating the eligible files to other constituent volumes within the FlexGroup. The storage system will experience some small performance degradation while the scanning process is running, so you may prefer to run your rebalance operations outside of peak storage workload operating hours in order to minimize this temporary performance impact.
- A rebalanced file experiences a small performance degradation that persists even after rebalancing completes, meaning that subsequent access to the rebalanced file will be fractionally slower than before the file was rebalanced. This is a consequence of the rebalance operation converting eligible file inodes to multipart inodes and redistributing the children inodes to different constituent volumes from the parent inode. The accompanying extra level of inode indirection required for joint file metadata and content access is the source of this degradation.
- The more files you rebalance on a FlexGroup volume, the more the aforementioned small, persistent performance degradations will accumulate, whether from an initial rebalance operation or from subsequent rebalance operations. Over time this growing accumulation can also affect the overall performance on the FlexGroup volume and not just the rebalanced files. If you rebalance wisely, meaning you limit the number of files that have undergone rebalancing in the FlexGroup, the performance gains that result from rebalancing

a seriously imbalanced FlexGroup outweigh these accumulated small performance drains. To maximize this likelihood, best practice is to rebalance infrequently, only when a storage imbalance is actually problematic. You should limit the number of files eligible for rebalancing by utilizing the rebalance operation's file size eligibility controls to restrict the size range of the files that will be candidates for rebalancing. Bear in mind that large files (100MB+) are typically the best rebalancing candidates.

In this lab activity you are a storage administrator performing an automatic rebalance of the newly expanded “balanceFG FlexGroup” volume. An automatic rebalance tends to be a long-running operation, with a duration that is dependant upon the size of the volume, the quantity and sizes of the files on the volume, the degree of imbalance, the client workloads that are using the volume, and so on. To keep this demonstration brief, the “balanceFG” FlexGroup volume uses substantially smaller constituent volumes than NetApp’s recommended minimum size of 100GB. The files on the “balanceFG” FlexGroup volume are also specifically designed so the automatic rebalance operation can complete in just a couple of minutes and produce clearly visible rebalancing results.

1. In your PuTTY session to “cluster1”, make sure that your CLI session is operating at the “diag” privilege level.

```
cluster1::> set -priv diag

Warning: These diagnostic commands are for use by NetApp personnel only.
Do you want to continue? {y|n}: y

cluster1::*>
```

2. The lab starts with a pre-created “balanceFG” FlexGroup volume already in existence. Query the constituent volume details for this FlexGroup volume. Also list the volume's snapshots.

```
cluster1::*> volume show -vserver svm1 -volume balanceFG* -fields size,available,used,percent-
used,files,files-used
vserver volume      size available used   percent-used files  files-used
-----
svm1    balanceFG 20GB 10.39GB 8.61GB 45%      622548 1043
svm1    balanceFG_0001 5GB 495.4MB 4.27GB 89%      155637 423
svm1    balanceFG_0002 9.24GB 4.50GB 4.28GB 48%      287759 427
svm1    balanceFG_0003 2.88GB 2.70GB 32.30MB 1%      155637 97
svm1    balanceFG_0004 2.88GB 2.70GB 32.29MB 1%      155637 96

5 entries were displayed.

cluster1::*> snap list -vserver svm1 -volume balanceFG
Vserver Volume Snapshot Size Total% Used%
-----
svm1    balanceFG base 8.43GB 42% 50%

cluster1::*>
```

This FlexGroup volume was initially 10 GB in size, comprised of two 5 GB constituent volumes. The FlexGroup volume was then populated with approximately 9 GB of file data, the “base” snapshot was created, and finally two additional 5 GB constituent volumes were added. The large difference in the “used” and “files-used” column values between the first two and last two constituent volumes shows that all of the “balanceFG” FlexGroup volume's files are currently hosted on the first two constituents.

You may observe the presence of other volumes snapshots besides “base” in your lab; these were generated by the volume's configured snapshot policy after the lab started.

3. In your PuTTY session to “rhel1”, examine the data set that has been pre-populated on the “balanceFG” FlexGroup volume.

```
[root@rhel1 ~] cd /svm1/balanceFG
[root@rhel1 balanceFG]# find . | wc -l
657
```

```
[root@rhell1 balanceFG]# ls
dir_0 dir_1 dir_2 dir_3 dir_4 dir_5 largedir_0 largedir_1
[root@rhell1 balanceFG]# ls dir_0
subdir_0 subdir_1 subdir_2 subdir_3
[root@rhell1 balanceFG]# ls -l dir_0/subdir_0
total 245300
-rw-r--r--. 1 root root 10002432 May 13 04:57 file_00.dat
-rw-r--r--. 1 root root 10002432 May 13 04:57 file_01.dat
.
# omitted for brevity
.
-rw-r--r--. 1 root root 10002432 May 13 04:58 file_23.dat
-rw-r--r--. 1 root root 10002432 May 13 04:58 file_24.dat
[root@rhell1 balanceFG]# ls -l largedir_0
total 1470672
-rw-r--r--. 1 root root 125001728 May 13 04:59 largefile_00.dat
-rw-r--r--. 1 root root 125001728 May 13 04:59 largefile_01.dat
.
# omitted for brevity
.
-rw-r--r--. 1 root root 125001728 May 13 05:00 largefile_10.dat
-rw-r--r--. 1 root root 125001728 May 13 05:00 largefile_11.dat
[root@rhell1 balanceFG]#
```

The “balanceFG” volume contains 657 items; the “balanceFG” directory itself, two largedir_* directories each containing twelve 125 GB binary data files, and six dir_* directories each containing four subdir_* subdirectories that in turn each hold twenty five 10 MB binary data files.

4. In your PuTTY session to “cluster1”, query the current rebalance status for the “balanceFG” FlexGroup volume.

```
cluster1::*> volume rebalance show -vserver svml -volume balanceFG*

Vserver: svml
```

Volume	State	Total	Used For Imbalance	Size Used by Snapshots	Target Used	Imbalance Size	%
balanceFG	not-running	20GB	8.55GB	340KB	-	4.21GB	49%
balanceFG__0001	not-running	7.21GB	4.23GB	156KB	2.14GB	2.10GB	98%
balanceFG__0002	not-running	7.25GB	4.25GB	184KB	2.14GB	2.11GB	98%
balanceFG__0003	not-running	2.77GB	32.28MB	0B	2.14GB	-2.11GB	-98%
balanceFG__0004	not-running	2.77GB	32.28MB	0B	2.14GB	-2.11GB	-98%

```
5 entries were displayed.
cluster1::*>
```

This command lists details for the “balanceFG” FlexGroup volume as a whole, as well as for each constituent volume. The actual values you see in your lab may differ a bit from what is shown in this example, this is normal.

The “state” column values indicate that no rebalance operation is currently in progress for this FlexGroup volume. The other reported metrics in this command output include:

- “Total”: The configured size of the volume.
- “Used For Imbalance”: The constituent volume’s consumed storage space, excluding snapshots.
- “Size Used by Snapshots”: The constituent volume’s consumed snapshot space.
- “Target Used”: The amount of space that would be consumed on the constituent volume if the FlexGroup volume’s overall contents were equally distributed across all of the constituent volumes. This is a theoretical ideal value used for level setting other metrics rather than an actual end result that you would likely ever achieve in the real world.
- “Imbalance Size”: How much the constituent’s volume’s current contents are greater than or less than the value in “Target Used”.

- “Imbalance Percentage”: The ratio of imbalance size to target size expressed as a percentage.



Tip: The “Used For Imbalance” column values reported in this command correspond fairly closely to the “used” column values you saw in the “volume show” command output you examined in step 2. This will not always be the case, as you will see in an upcoming step.

As you would expect for a FlexGroup volume with newly added constituents, the overall “Imbalance %” for this FlexGroup is 49%, which is pretty high. The listed metrics report that the older constituents are substantially fuller than the newer ones are, making this FlexGroup volume a potential candidate for a rebalance operation.

5. Enable multipart inode granular data functionality on the “balanceFG” FlexGroup volume, which is a required to perform a rebalance operation for FlexGroup.

```
cluster1::*> volume show -vserver svm1 -volume balanceFG -fields granular-data
vserver volume      granular-data
-----
svm1      balanceFG disabled

cluster1::*> volume modify -vserver svm1 -volume balanceFG -granular-data enabled

Warning: Granular data will be enabled for volume "balanceFG" in Vserver
"svm1". Once enabled, this property can only be disabled by restoring
a Snapshot copy. Earlier versions of ONTAP are not compatible with
this property, therefore revert to an earlier version of ONTAP will
not be possible unless this volume is deleted or restored to a
Snapshot copy that was taken before this property was enabled.
Do you want to continue? {y|n}: y
[Job 376] Job succeeded: volume modify succeeded

cluster1::*> volume show -vserver svm1 -volume balanceFG -fields granular-data
vserver volume      granular-data
-----
svm1      balanceFG enabled

cluster1::*>
```

Multipart inode granular data functionality is disabled by default for FlexGroup volumes. You can enable this functionality at volume creation time, or later on as is shown here. If you try to start a rebalance on a FlexGroup volume that does not have granular data enabled, that command will ask whether you want to enable granular data at that time.

6. Start a rebalance operation for the “balanceFG” FlexGroup volume.

```
cluster1::*> volume rebalance start -vserver svm1 -volume balanceFG

Error: command failed: Failed to start volume capacity rebalancing for volume
"balanceFG" in Vserver "svm1". Reason: The next scheduled Snapshot copy
for volume "balanceFG" in Vserver "svm1" is scheduled for Sat May 13
21:05:00 2023, in 0h10m28s. This is within the specified 6h0m0s maximum
runtime for the volume capacity rebalancing operation. To run the
operation, either reduce the "-max-runtime" or disable the Snapshot
policy for the volume.

cluster1::*>
```

ONTAP has detected that a snapshot is scheduled to take place for the volume during the command’s scheduled six hour run time window. Since ONTAP does not permit the creation of snapshots on a FlexGroup volume that is undergoing rebalance, the rebalance command fails. There are two options for resolving this issue:

- Reduce the length of the rebalance operation’s run time window.
- Disable snapshots for FlexGroup volume.

7. Examine the rebalance operation configuration settings for the “balanceFG” FlexGroup volume.

```
cluster1::*> volume rebalance show -vserver svm1 -volume balanceFG -config

Vserver: svm1
```

Volume	Max Runtime	Threshold Min	Threshold Max	Max File Moves	Min File Size	Exclude Snapshot
balanceFG	6h0m0s	5%	20%	25	100MB	true

```
cluster1:*>
```

The rebalance scanning process uses these configuration settings to control rebalancing operation behavior. A description of each setting follows:

- “Max Runtime”: Specifies the maximum time a capacity rebalancing operation will run. The default value is 6 hours.
- “Threshold Min”: When a constituent’s imbalance percentage is smaller than this value, files will not be moved from the constituent. The default value is 5%.
- “Threshold Max”: When a constituent’s imbalance percentage is larger than this value, files will be moved from the constituent. The default value is 20%.
- “Max File Moves”: Specifies the maximum number of file moves that a volume capacity rebalancing operation can run concurrently on a constituent of the FlexGroup volume. The default value is 25.
- “Min File Size”: Specifies the minimum file size a file must exceed in order to be eligible for volume capacity rebalancing. The default value is 100MB, and this value must be a multiple of 4KB. Setting the minimum file size to less than the default value typically leads to more for the FlexGroup’s files being moved.
- “Exclude Snapshot”: Specifies whether files with blocks captured in snapshots should be excluded (i.e., considered ineligible) in a volume capacity rebalancing operation. The default value is true.

These settings are scoped to the FlexGroup volume, so any changes you make to them will apply to all future rebalancing operations for this specific FlexGroup volume. You can also override these values on a one-time basis by including appropriate CLI options to the “volume rebalance start” command.



Tip: The “Min File Size” setting is your primary control knob for managing file rebalancing eligibility. Lowering this file size typically results in more of the FlexGroup volume’s files being eligible for rebalance. Since your goal should be to rebalance as few files as necessary to alleviate a problematic imbalance (see the introduction of this lab activity for more details), and large files tend to produce the greatest space rebalance benefit at the lowest overall file count cost, a conservative rebalancing strategy is to specify a size value that is at greater than or equal to the 100MB default.

8. Modify the rebalance operation configuration settings for the balanceFG FlexGroup volume to reduce the maximum runtime window, and also make files in snapshots eligible for rebalancing.

```
cluster1:*> volume rebalance modify -vsrv svml -volume balanceFG -max-runtime 30m -exclude-
snapshots false

cluster1:*> volume rebalance show -vsrv svml -volume balanceFG -config

Vserver: svml
Volume      Max Runtime  Threshold Min  Max  Max File Moves  Min File Size  Exclude Snapshot
-----
balanceFG   0h30m0s     5%    20%           25      100MB      false

cluster1:*>
```

By default a rebalance operation excludes otherwise eligible files that have blocks that are captured in snapshots. Setting “-exclude-snapshots false” removes this restriction. A rebalance of such a file is actually implemented as a copy of the locked blocks rather than a move since those blocks are still locked in the snapshot on the source constituent, and must continue to reside there (consuming space) until the snapshot is deleted.

Remember that a running rebalance scan does impose a small performance penalty on the storage system. As balanceFG is much smaller than typically recommended for a FlexGroup volume, and the dataset it contains is designed to rebalance in just a couple of minutes, the default 6 hour value for “Max Runtime” is rather excessive for this lab activity. 30 minutes is the smallest maximum runtime window value that ONTAP allows.

Reducing the value of “Max Runtime” is not the most effective strategy for working around the upcoming scheduled snapshot problem you encountered earlier. This is because the default snapshot policy used by the balanceFG FlexGroup volume calls for creating a snapshot at five minutes past every hour, and chances are high that the next snapshot will trigger within the next 30 minutes. The more dependable strategy is to disable snapshots for the FlexGroup volume.

- Query ONTAP for the balanceFG FlexGroup volume's current snapshot policy, then disable snapshots for this FlexGroup.

```
cluster1::*> volume show -vserver svm1 -volume balanceFG -fields snapshot-policy
vserver volume      snapshot-policy
-----
svm1      balanceFG default

cluster1::*> volume modify -vserver svm1 -volume balanceFG -snapshot-policy none

Warning: You are changing the Snapshot policy on volume "balanceFG" to "none".
        Snapshot copies on this volume that do not match any of the prefixes
        of the new Snapshot policy will not be deleted. However, when the new
        Snapshot policy takes effect, depending on the new retention count,
        any existing Snapshot copies that continue to use the same prefixes
        might be deleted. See the 'volume modify' man page for more
        information.
Do you want to continue? {y|n}: y
[Job 377] Job succeeded: volume modify succeeded

cluster1::*> volume show -vserver svm1 -volume balanceFG -fields snapshot-policy
vserver volume      snapshot-policy
-----
svm1      balanceFG none

cluster1::*>
```



Note: You will need to re-apply the previous snapshot policy to this FlexGroup volume again after the rebalance operation completes.



Tip: Data Protection policies also use snapshots. If this FlexGroup volume was configured with a data protection policy (so that SnapMirror would replicate its contents to another volume) then that data protection policy would likely also need to be disabled before you could start a rebalance operation for this FlexGroup volume.

- Re-query the rebalance metrics for the balanceFG FlexGroup volume so those values are conveniently available.

```
cluster1::*> volume rebalance show -vserver svm1 -volume balanceFG*

Vserver: svm1
```

Volume	State	Total	Used For Imbalance	Size Used by Snapshots	Target Used	Imbalance Size	%
balanceFG	not-running	20GB	8.55GB	340KB	-	4.21GB	49%
balanceFG__0001	not-running	7.21GB	4.23GB	156KB	2.14GB	2.10GB	98%
balanceFG__0002	not-running	7.25GB	4.25GB	184KB	2.14GB	2.11GB	98%
balanceFG__0003	not-running	2.77GB	32.28MB	0B	2.14GB	-2.11GB	-98%
balanceFG__0004	not-running	2.77GB	32.28MB	0B	2.14GB	-2.11GB	-98%

5 entries were displayed.

```
cluster1::*>
```

The balanceFG FlexGroup's reported "Used For Imbalance" overall value is approximately 8.5 GB, and its "Size Used by Snapshots" is just a few KB or MB. You will re-examine these values again after rebalancing.

- Start a new rebalance operation for the balanceFG FlexGroup.

```
cluster1::*> volume rebalance start -vserver svm1 -volume balanceFG

cluster1::*>
```

12. Periodically (every 10-15 seconds) query the status of the rebalance operation. In this lab activity the actual file moves usually complete within a minute or two of issuing the rebalance start command, although the overall rebalance operation continues scanning since the 30 minute runtime window remains in effect.



Note: The timing of the rebalancing activity in your lab session may differ from what is shown in this example.

```
cluster1::*> # This time point corresponds to immediately after issuing the rebalance command...

cluster1::*> volume rebalance show -vserver svm1 -volume balanceFG*

Vserver: svm1
```

Volume	State	Total	Used For Imbalance	Size Used by Snapshots	Target Used	Imbalance Size	%
balanceFG	idle	20GB	8.55GB	340KB	-	4.21GB	49%
balanceFG__0001	idle	7.21GB	4.23GB	156KB	2.14GB	2.10GB	98%
balanceFG__0002	idle	7.25GB	4.25GB	184KB	2.14GB	2.11GB	98%
balanceFG__0003	idle	2.77GB	32.28MB	0B	2.14GB	-2.11GB	-98%
balanceFG__0004	idle	2.77GB	32.28MB	0B	2.14GB	-2.11GB	-98%

5 entries were displayed.

```
cluster1::*> # This time point corresponds to approximately 10 seconds after issuing the rebalance command...

cluster1::*> volume rebalance show -vserver svm1 -volume balanceFG*

Vserver: svm1
```

Volume	State	Total	Used For Imbalance	Size Used by Snapshots	Target Used	Imbalance Size	%
balanceFG	rebalancing	20GB	8.55GB	340KB	-	4.21GB	49%
balanceFG__0001	scanning	7.21GB	4.23GB	156KB	2.14GB	2.10GB	98%
balanceFG__0002	scanning	7.25GB	4.25GB	184KB	2.14GB	2.11GB	98%
balanceFG__0003	idle	2.77GB	32.28MB	0B	2.14GB	-2.11GB	-98%
balanceFG__0004	idle	2.77GB	32.28MB	0B	2.14GB	-2.11GB	-98%

5 entries were displayed.

```
cluster1::*> # This time point corresponds to approximately 20 seconds after issuing the rebalance command...

cluster1::*> volume rebalance show -vserver svm1 -volume balanceFG*

Vserver: svm1
```

Volume	State	Total	Used For Imbalance	Size Used by Snapshots	Target Used	Imbalance Size	%
balanceFG	rebalancing	20GB	8.90GB	424KB	-	3.80GB	42%
balanceFG__0001	rebalancing-source	7.21GB	4.12GB	212KB	2.22GB	1.89GB	85%
balanceFG__0002	rebalancing-source	7.25GB	4.22GB	89.48MB	2.28GB	1.95GB	85%
balanceFG__0003	rebalancing-dest	2.77GB	391.4MB	0B	2.22GB	-1.84GB	-82%
balanceFG__0004	rebalancing-dest	2.77GB	271.7MB	0B	2.28GB	-2.01GB	-88%

5 entries were displayed.

```
cluster1::*> # This time point corresponds to approximately 40 seconds after issuing the rebalance command...

cluster1::*> volume rebalance show -vserver svm1 -volume balanceFG*

Vserver: svm1
```

Volume	State	Total	Used For Imbalance	Size Used by Snapshots	Target Used	Imbalance Size	%
balanceFG	rebalancing	20GB	8.90GB	424KB	-	3.80GB	42%
balanceFG__0001	rebalancing-source	7.21GB	4.12GB	212KB	2.22GB	1.89GB	85%
balanceFG__0002	rebalancing-source	7.25GB	4.22GB	89.48MB	2.28GB	1.95GB	85%
balanceFG__0003	rebalancing-dest	2.77GB	391.4MB	0B	2.22GB	-1.84GB	-82%
balanceFG__0004	rebalancing-dest	2.77GB	271.7MB	0B	2.28GB	-2.01GB	-88%

5 entries were displayed.

```

Volume      State      Total  Imbalance  Used by  Target  Imbalance
-----  -----  -----  -----  -----  -----  -----
balanceFG   rebalancing  20GB    9.13GB    2.11GB    -    1.70GB    18%
balanceFG__0001 rebalancing-source  7.21GB    3.18GB    990.8MB    2.28GB    922.3MB    39%
balanceFG__0002 rebalancing-source  7.25GB    3.08GB    1.14GB    2.28GB    820.9MB    35%
balanceFG__0003 rebalancing-dest  2.77GB    1.43GB    0B    2.28GB    -875.5MB    -37%
balanceFG__0004 rebalancing-dest  2.77GB    1.43GB    0B    2.28GB    -867.8MB    -37%
5 entries were displayed.

cluster1::*> # This time point corresponds to approximately 80 seconds after issuing the rebalance
command...

cluster1::*> volume rebalance show -vserver svml -volume balanceFG*

Vserver: svml
Volume      State      Total  Imbalance  Used by  Target  Imbalance
-----  -----  -----  -----  -----  -----  -----
balanceFG   rebalancing  20GB    8.55GB    2.53GB    -    1.41GB    16%
balanceFG__0001 scanning  6.34GB    2.83GB    1.23GB    2.14GB    710.1MB    32%
balanceFG__0002 scanning  6.37GB    2.85GB    1.30GB    2.14GB    730.2MB    33%
balanceFG__0003 idle  2.77GB    1.43GB    0B    2.14GB    -720.2MB    -32%
balanceFG__0004 idle  4.51GB    1.43GB    0B    2.14GB    -720.1MB    -32%
5 entries were displayed.

cluster1::*>

```

As you iterate this command, you should see the state of the individual constituent volumes change from “not-running” to possible values of “idle”, “scanning”, “rebalancing” (including “rebalancing-source” and “rebalancing-dest”), and then eventually to either “scanning” or “idle” again. (The state of the overall “balanceFG” entry changes to “rebalancing” for the duration of the scanning window).

As rebalancing proceeds, each constituent’s “Imbalance Size” and “Imbalance %” metrics start converging towards more similar values. The values will never reach true convergence, as many of balanceFG’s files are too small to be eligible for rebalancing. Even if all of balanceFG’s files were large enough to be eligible, you would still not achieve true convergence as the “Threshold Min” and “Threshold Max” rebalance configuration setting values you saw in step 7 would intervene to prevent that outcome.

For this particular FlexGroup volume and dataset, the overall “Imbalance %” typically settles at around 16%. Individual constituents settle at around +/- 32% but you may see a somewhat less uniform range in your lab than is shown in this example.

13. Stop the rebalance operation and then re-query the rebalance status to confirm that rebalancing has stopped.

```

cluster1::*> volume rebalance stop -vserver svml -volume balanceFG

cluster1::*> volume rebalance show -vserver svml -volume balanceFG*

Vserver: svml
Volume      State      Total  Imbalance  Used by  Target  Imbalance
-----  -----  -----  -----  -----  -----  -----
balanceFG   not-running  20GB    8.55GB    2.76GB    -    1.41GB    16%
balanceFG__0001 not-running  6.32GB    2.83GB    1.38GB    2.14GB    710.3MB    32%
balanceFG__0002 not-running  6.34GB    2.85GB    1.38GB    2.14GB    729.4MB    33%
balanceFG__0003 not-running  3.67GB    1.43GB    0B    2.14GB    -719.9MB    -32%
balanceFG__0004 not-running  3.67GB    1.43GB    0B    2.14GB    -719.8MB    -32%
5 entries were displayed.

```

```
cluster1::*>
```

Take a look at the FlexGroup's overall "Size Used by Snapshots" metric, which has shot up to approximately 2.7 GB from the original few KB/MB value you saw in step 10. The first two constituents account for nearly all of that snapshot space consumption, representing the blocks locked in snapshots that were left behind as eligible files were rebalanced to other constituents. Significantly, the FlexGroup's overall "Used For Imbalance" value remains unchanged at approximately 8.5 GB.

14. Query the constituent volume details for the balanceFG FlexGroup volume.

```
cluster1::*> volume show -vserver svml -volume balanceFG* -fields used,size,percent-
used,files,files-used
vserver volume      size used    percent-used files  files-used
-----
svml     balanceFG 20GB 10.79GB 56%      823404 1077
svml     balanceFG__0001 6.32GB 3.95GB 65%      224371 424
svml     balanceFG__0002 6.34GB 3.97GB 65%      287759 429
svml     balanceFG__0003 3.67GB 1.43GB 41%      155637 112
svml     balanceFG__0004 3.67GB 1.43GB 41%      155637 112
5 entries were displayed.
cluster1::*>
```

The "used" for the new and old constituents are much closer together than before. The FlexGroup's overall "used" space reports as approximately 10.8GB. This is the FlexGroup volume's true space consumption, including active files and snapshots.

15. Delete all the snapshots from the balanceFG FlexGroup volume. The list of snapshots you are prompted to delete may vary from those shown in the following example.

```
cluster1::*> snapshot delete -vserver svml -volume balanceFG -snapshot *

Warning: Deleting a Snapshot copy permanently removes data that is stored only
in that Snapshot copy. Are you sure you want to delete Snapshot copy
"base" for volume "balanceFG" in Vserver "svml" ? {y|n}: y
1 entry was acted on.
cluster1::*>
```

16. Query the rebalance operation status one last time.

```
cluster1::*> volume rebalance show -volume balanceFG*

Vserver: svml

Volume      State      Total    Used For Imbalance    Size Used by Snapshots    Target Used    Imbalance Size    %
-----
balanceFG  not-running    20GB    8.55GB    0B    -    1.41GB    16%
balanceFG__0001  not-running    5GB    2.83GB    0B    2.14GB    710.3MB    32%
balanceFG__0002  not-running    5.01GB    2.85GB    0B    2.14GB    729.4MB    33%
balanceFG__0003  not-running    5GB    1.43GB    0B    2.14GB    -719.9MB    -32%
balanceFG__0004  not-running    4.99GB    1.43GB    0B    2.14GB    -719.8MB    -32%
5 entries were displayed.
cluster1::*>
```

The values under the "Files Used by Snapshots" column should have dropped to zero.

17. Query the constituent volume details for the balanceFG FlexGroup volume.

```
cluster1::*> volume show -vserver svml -volume balanceFG* -fields used,size,percent-used,files,files-used
vserver volume      size used    percent-used files  files-used
-----
svml    balanceFG 20GB 8.60GB 45%      864501 1077
svml    balanceFG_0001
          5GB 2.85GB 60%      224371 424
svml    balanceFG_0002
          5.01GB
          2.87GB 60%      287759 429
svml    balanceFG_0003
          5GB 1.43GB 30%      196734 112
svml    balanceFG_0004
          4.99GB
          1.43GB 30%      155637 112
5 entries were displayed.
cluster1::*>
```

The reported “used” space should start dropping as ONTAP releases blocks that were previously locked in the snapshots you just deleted. That process is not instantaneous, so you may need to re-issue the command a couple of times until ONTAP finishes the releases, which you will know has happened once balanceFG’s “used” value reaches approximately 8.6 GB.

18. Re-apply the balanceFG FlexGroup volume’s previous snapshot policy so that automatic snapshot creation will resume for the balanceFG FlexGroup volume.

```
cluster1::*> volume modify -vserver svml -volume balanceFG -snapshot-policy default

Warning: You are changing the Snapshot policy on volume "balanceFG" to "default".
Snapshot copies on this volume that do not match any of the prefixes
of the new Snapshot policy will not be deleted. However, when the new
Snapshot policy takes effect, depending on the new retention count,
any existing Snapshot copies that continue to use the same prefixes
might be deleted. See the 'volume modify' man page for more
information.
Do you want to continue? {y|n}: y
[Job 402] Job succeeded: volume modify succeeded

cluster1::*> volume show -vserver svml -volume balanceFG -fields snapshot-policy
vserver volume      snapshot-policy
-----
svml    balanceFG default

cluster1::*>
```

19. Examine detailed metrics for the just completed rebalancing operation.

```
cluster1::*> volume rebalance show -vserver svml -volume balanceFG

Vserver Name: svml
Volume Name: balanceFG
Is Constituent: false
Rebalance State: not-running
Rebalance Notice Messages: -
Total Size: 20GB
Used Size for Calculating Imbalance: 8.55GB
Volume Size Used by Snapshot Copies: 0B
Constituent Target Used Size: -
Imbalance Size: 1.41GB
Imbalance Percentage: 16%
Moved Data Size: -
Maximum Constituent Imbalance Percentage: 33%
Rebalance Start Time: Sat May 13 22:34:16 2023
Rebalance Stop Time: Sat May 13 22:36:05 2023
Rebalance Runtime: 0h1m49s
Rebalance Maximum Runtime: 0h30m0s
Maximum Imbalance Threshold per Constituent: 20%
Minimum Imbalance Threshold per Constituent: 5%
Maximum Concurrent File Moves per Constituent: 25
Minimum File Size: 100MB
Exclude Files Stuck in Snapshot Copies: false
Most Recent File Move Timestamp: Sat May 13 22:34:44 2023
File Moves Started: 12
```

```

Files Scanned: 2951
Small Files Skipped: 2691
Large Files Skipped: 0
Fast Truncate Files Skipped: 0
Files in Snapshot Copies Skipped: 0
Files Skipped Due to Efficiency (by Block Count): 0
Files Skipped Due to Efficiency (by Percent): 0
Incompatible Files Skipped: 152
Metadata Files Skipped: 96
Remote Caches Skipped: 0
Write Fenced Files Skipped: 0
On Demand Destination Files Skipped: 0
Invalid Footprint Files Skipped: 0
All Other Files Skipped: 0
Blocks Scanned: 6969027
Blocks Skipped Due to Small Files: 6601023
Blocks Skipped Due to Large Files: 0
Blocks Skipped Due to Fast Truncate Files: 0
Blocks Skipped Due to Files in Snapshot Copies: 0
Blocks Skipped Due to Efficiency (by Block Count): 0
Blocks Skipped Due to Efficiency (by Percent): 0
Blocks Skipped Due to Incompatible Files: 144
Blocks Skipped Due to Metadata Files: 192
Blocks Skipped Due to Remote Caches: 0
Blocks Skipped Due to Write Fenced Files: 0
Blocks Skipped Due to On Demand Destination Files: 0
Blocks Skipped Due to Invalid Footprint Files: 0
Blocks Skipped Due to All Other Reasons: 0
Last File Move Error Timestamp: -
Last File Move Error File ID: -
Last File Move Error Destination Constituent DSID: -
Last File Move Error Code: -

cluster1::*>

```

The “File Moves Started” field lists the number of eligible files that the rebalancing operation relocated. The value reported in this command is incorrect and appears to be the result of an ONTAP bug. You can work around the bug by adding up the “File Moves Started” metrics for each constituent volume.

20. Display the “File Moves Started” metric for each constituent.

```

cluster1::*> volume rebalance show -vserver svml -volume balanceFG* -fields file-moves-started -is-constituent true
vserver volume          file-moves-started
-----
svml      balanceFG__0001 12
svml      balanceFG__0002 12
svml      balanceFG__0003 0
svml      balanceFG__0004 0
4 entries were displayed.

cluster1::*>

```

The total of these values is 24, the same number of files larger than 100 MB that are present on balanceFG. In summary, the rebalance operation moved all of those files to the newer constituents, while all 600 of the remaining (smaller) files remain in place on the older constituents.

Takeaway: When FlexGroup constituent imbalances become problematic, a storage administrator can work to non-disruptively improve FlexGroup volume performance by initiating an automatic rebalance operation to redistribute a selectable subset of the FlexGroup’s existing files across its constituent volumes.

2.5 FlexGroup Volume Move

ONTAP provides a method to non-disruptively move volumes between aggregates, perhaps because you have a cluster node or aggregate that is being over-utilized, or because you need to move the volume from a performance tier to a capacity tier or vice-versa. This capability also applies to FlexGroup volumes, specifically at the constituent volume level.

 **Tip:** For best results, NetApp recommends that you host all of a FlexGroup’s constituent volumes on the same performance tier. If you need to move a FlexGroup volume to a different performance tier then you should work to move all of the constituents volumes within a reasonably short time frame.

In this lab activity a storage administrator needs to relocate a constituent volume to a different aggregate because the current aggregate is running out of space.

1. This lab activity requires PuTTY sessions for “rhel1” and “cluster”. If you need to open a PuTTY session for either of these hosts, on the task bar of Jumpshot, right-click the **PuTTY icon** and select **PuTTY** from the context menu.

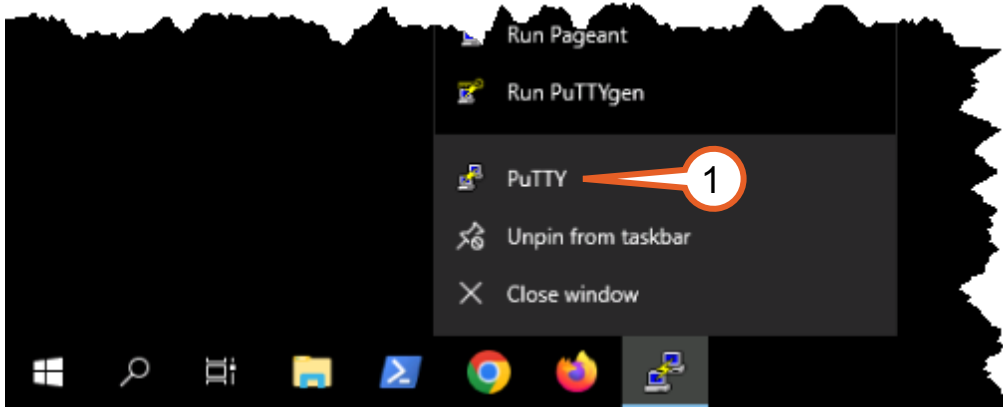


Figure 2-8:

The “PuTTY Configuration” dialog opens.

2. Double-click the session name for **your desired host** in the “Saved Sessions” list.

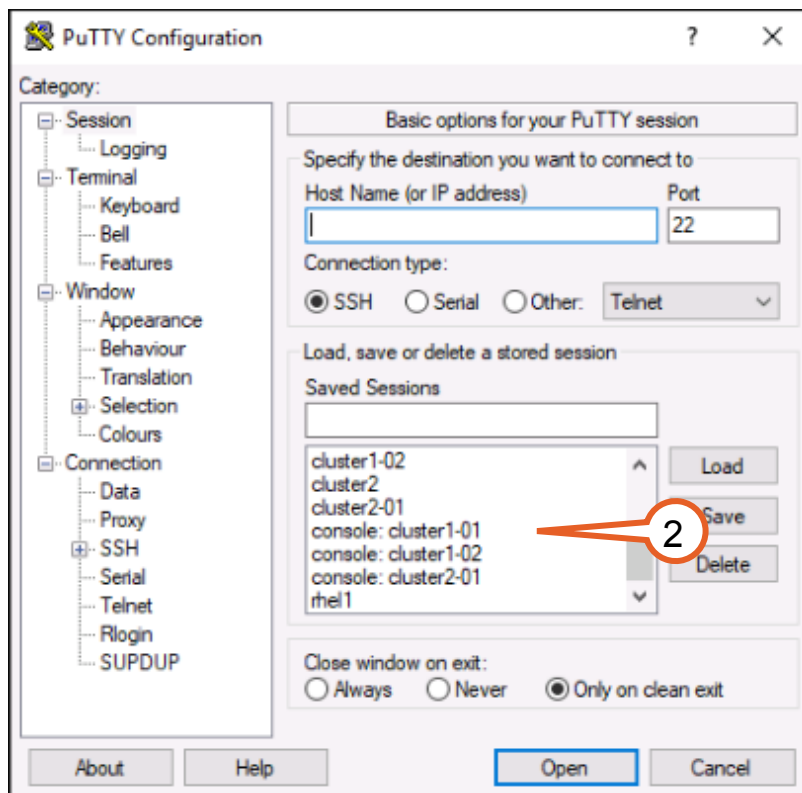


Figure 2-9:

3. Log in using the following credentials:
 - Username for cluster1: `admin`
 - Username for rhel1: `root`
 - Password: `Netapp1!`

4. Arrange these PuTTY session windows on your desktop so you can simultaneously view the full contents of both windows.
5. In your PuTTY session for “cluster1”, make sure your CLI session is operating at the “diag” privilege level and then examine the “demoFG” FlexGroup’s constituent volumes and current capacity metrics.

```
cluster1::*> set -priv diag

cluster1::*> volume show -vserver svml -volume demoFG*
Vserver   Volume           Aggregate          State    Type    Size    Available    Used%
-----
svml      demoFG           -                  online   RW       200GB    186.8GB     1%
svml      demoFG__0001     aggr1_cluster1_01 online   RW       100GB    93.39GB     1%
svml      demoFG__0002     aggr1_cluster1_02 online   RW       100GB    93.39GB     1%
3 entries were displayed.

cluster1::*> volume show -vserver svml -volume demoFG* -fields used,size,percent-used,files,files-
used,aggregate
vserver volume aggregate size used percent-used files files-used
-----
svml      demoFG -          200GB 3.21GB 1%          6225918 22319
svml      demoFG__0001     aggr1_cluster1_01 100GB 1.61GB 1%          3112959 11155
svml      demoFG__0002     aggr1_cluster1_02 100GB 1.61GB 1%          3112959 11164
3 entries were displayed.

cluster1::*>
```

The demoFG FlexGroup is comprised of two 100 GB constituent volumes, each containing about 1.6 GB of used space. Observe that the demoFG__0001 constituent volume currently resides on the aggr1_cluster1_01 aggregate.

The dataset on the demoFG FlexGroup volume consists of 20,000 small text files and sixteen 200 MB binary files distributed across approximately 2200 directories and subdirectories.

6. Queue up the following command that you will use to initiate the volume move *but don’t execute it yet*.

```
volume move start -vserver svml -volume demoFG__0001 -destination-aggregate aggr1_cluster1_02
```

When you execute this command in an upcoming step of this lab activity, it will initiate a volume move of the demoFG__0001 constituent volume from its current home on the aggr1_cluster1_01 aggregate to the aggr1_cluster1_02 aggregate.

7. In your PuTTY session to “rhel1”, initiate an I/O workload on the demoFG volume and immediately proceed to the next step without waiting for the script to finish. This script helps demonstrate that the upcoming volume move is nondisruptive, as it will overwrite all 20,000 of the volume’s existing small text files with new contents while the move is happening. This script typically completes in about 40 seconds.

```
[root@rhel1 flexgroups]# cd ~/flexgroups
[root@rhel1 flexgroups]# ./populate_demoFG.sh

#####
##### Starting File Population for #####
##### /svml/demoFG #####
#####

#####
##### Populating small text files
#####
...
```

8. In your PuTTY session to “cluster1”, execute the volume move command you queued up in step 6, and then query the move operation status until it completes. In this lab activity the move typically completes within about 30 seconds.

```
cluster1::*> volume move start -vserver svml -volume demoFG__0001 -destination-aggregate
aggr1_cluster1_02
```

```
[Job 190] Job is queued: Move "demoFG__0001" in Vserver "svml" to aggregate "aggr1_cluster1_02".
Use the "volume move show -vserver svml -volume demoFG__0001" command to view the status of this
operation.
```

```
cluster1:*> vol move show -vserver svml
Vserver   Volume      State      Move Phase  Percent-Complete  Time-To-Complete
-----
svml      demoFG__0001
          healthy   replicating
          14%
          Sat May 13 19:47:25 2023

cluster1:*> volume move show -vserver svml
Vserver   Volume      State      Move Phase  Percent-Complete  Time-To-Complete
-----
svml      demoFG__0001
          done      completed  100%
          -

cluster1:*>
```

Provided that you initiate the volume move right after launching the populate_demoFG.sh script, the move should complete before the script does.

9. View greater detail of the just completed volume move operation. This guide won't delve into the specifics of these fields.

```
cluster1:*> volume move show -vserver svml -volume demoFG__0001

Vserver Name: svml
Volume Name: demoFG__0001
Volume Instance UUID: 7cf27827-e529-11ed-80c3-005056b011ab
Actual Completion Time: Sat May 13 19:47:42 2023
Bytes Remaining: -
Bytes Sent: -
Status Code: 0
Completion String: Successful
Specified Action For Cutover: retry_on_failure
Specified Cutover Attempts: 3
Specified Cutover Time Window: 30
Time User Triggered Cutover: -
Time Move Job Last Entered Cutover: Sat May 13 19:47:31 2023
Cutovers Attempted: 1
Times Cutover Hard Deferred: 0
Times Cutover Soft Deferred: 0
Destination Aggregate: aggr1_cluster1_02
Destination Node: cluster1-02
Detailed Status: Successful
Estimated Time of Completion: -
Internal Progress of Move: -
Actual State of Job: -
Job ID: 190
Job UUID: f8c6dd26-f1c6-11ed-8e95-005056b0b1b5
Managing Node: cluster1-01
Percentage Complete: 100%
Move Phase: completed
Prior Issues Encountered: -
Estimated Remaining Duration: -
Replication Throughput: -
Duration of Move: 00:00:23
Source Aggregate: aggr1_cluster1_01
Source Node: cluster1-01
Start Time of Move: Sat May 13 19:47:19 2023
Move State: done
Move Initiated by Auto Balance Aggregate: false
Bypass Replication Engine Throttling: false
Skip the Delta Calculation: false
Time Taken to Complete Cutover: 1s
Original Job ID: -
Is Source Volume Encrypted: false
Encryption Key ID of Source Volume:
Is Destination Volume Encrypted: false
Encryption Key ID of Destination Volume:
Enforce Replication Engine Node Level Network Throttling: false

cluster1:*>
```

10. In your PuTTY session to “rhel1”, examine the output of the populate_demoFG.sh script. If this script had encountered a write error during the move then it would have generated an exception. No error is indicated.



Note: You do not actually run this command in this step, you just look at the output of the command you ran in step 7.

```
[root@rhel1 flexgroups]# ./populate_demoFG.sh

#####
##### Starting File Population for #####
##### /svml/demoFG #####
#####

####
#### Populating small text files
####

Overlaying script defaults using /root/flexgroups/demoFG-text.json.

Populating /svml/demoFG with the following:
  100 directories (dir_N).
  20 subdirectories (subdir_N) per directory.
  10 random text files (file_N.txt, 6-40 bytes) per subdirectory.
  70% of all created files will be unique.
  Files that already exist will be overwritten.
  In total, up to 20,000 files will potentially be distributed across 2,101 directories.

Population task starting now --->
.....
<--- Population task complete

Created/updated 20,000 out of 20,000 potential file candidates
Population task elapsed time = 50.816 seconds

####
#### Populating large binary files
####

Overlaying script defaults using /root/flexgroups/demoFG-binary.json.

Populating /svml/demoFG with the following:
  4 directories (largedir_N).
  4 random binary files (largefile_N.dat, 200MiB bytes) per directory.
  100% of all created files will be unique.
  Files that already exist will not be overwritten.
  In total, up to 16 files will potentially be distributed across 5 directories.

Population task starting now --->
....
<--- Population task complete

Created/updated 0 out of 16 potential file candidates
Info: All files were pre-existing and no files were overwritten!
Population task elapsed time = 0.007 seconds

#####
##### File Population Complete for #####
##### /svml/demoFG #####
##### Total Time = 51 seconds #####
#####

[root@rhel1 flexgroups]#
```

11. In your PuTTY session to “cluster1”, verify that both of demoFG’s constituent volumes are now resident on the aggr1_cluster1_02 aggregate.

```
cluster1::*> volume show -vserver svml -volume demoFG*
Vserver   Volume          Aggregate      State    Type    Size    Available Used%
-----
svml      demoFG          -             online   RW      200GB   186.4GB  1%
svml      demoFG__0001    aggr1_cluster1_02 online   RW      100GB   93.19GB  1%
svml      demoFG__0002    aggr1_cluster1_02 online   RW      100GB   93.19GB  1%

3 entries were displayed.

cluster1::*>
```

12. **Takeaway:** Storage administrators can non-disruptively migrate member volumes to a different node and aggregates in response to changing storage management requirements.

2.6 Use SnapMirror to Restore a Single File

ONTAP snapshots capture the read-only, point-in-time state of a volume, hosted locally within the volume's own storage space. The captured state only records changes made to the volume's files since the last snapshot was taken, and restoring a file from a snapshot is fast and easy because both the restore source and destination are local to the volume. Snapshots typically consume minimal storage space as the rate of content change of most volumes tends to be relatively low, enough so that most customers are able to maintain a sizeable number of snapshots on the local volume. This provides a good selection of point-in-time options to choose from for file restore operations.

If a volume exhibits a high rate of content change then local snapshots consume much more space, perhaps enough to significantly interfere with the volume's primary purpose of supporting a production workload. Simply increasing the size of the volume to compensate may not be cost effective, especially if the volume utilizes expensive high performance flash. An alternative protection solution is to use NetApp SnapMirror to replicate those captured snapshots to another volume that utilizes lower cost storage technology like spinning disks. This allows you to maintain a much large selection of snapshot history on the destination volume, long after those snapshots no longer exist on the source volume. That destination volume can also reside on a completely separate ONTAP cluster, protecting you against data loss in the event of a disaster scenario affecting your primary storage.

While you might assume that restoring a file from one cluster to another would be complicated, ONTAP allows a storage administrator to restore a single file by running one CLI command on the primary storage cluster.

In this lab activity, a storage administrator wants to restore a version of a file that is no longer available in local snapshots. Fortunately, the organization uses NetApp SnapMirror to maintain a longer archive of the volume's snapshots on another ONTAP cluster, so the storage administrator simply needs to restore the file back to the primary cluster source volume from that secondary cluster.

1. This lab activity requires PuTTY sessions for "rhel1", "cluster1", and "cluster2", and you will be frequently switching between them. If you need to open a PuTTY session for any of these hosts, on the task bar of Jumphost, right-click the **PuTTY icon** and select **PuTTY** from the context menu.

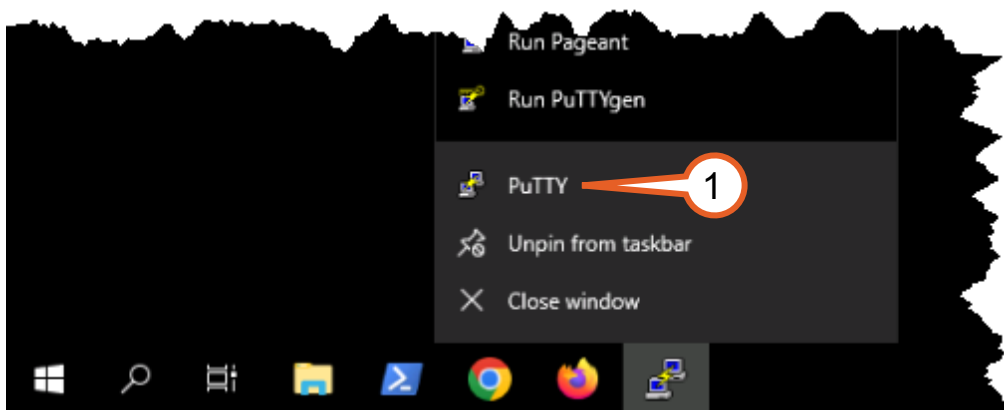


Figure 2-10:

The "PuTTY Configuration" dialog opens.

- Double-click the session name for **your desired host** in the “Saved Sessions” list.

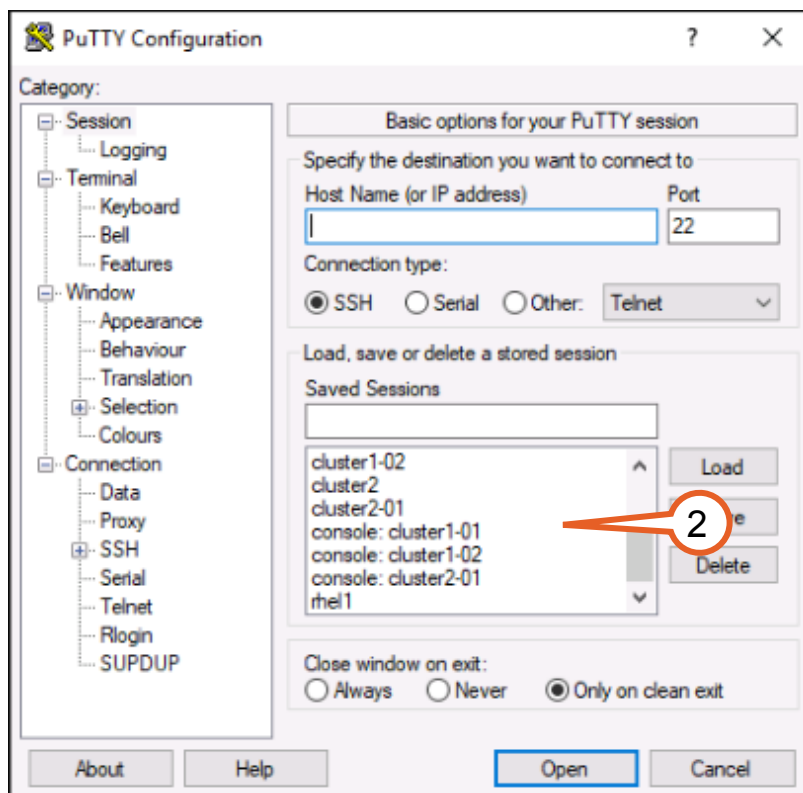


Figure 2-11:

- Log in using the following credentials:

- Username (for cluster1 or cluster2): **admin**
- Username (for rhel1): **root**
- Password: **Netapp1!**

On your display, arrange the PuTTY windows for “cluster1” and “cluster2” so you can simultaneously view the full contents of both windows. Maintaining simultaneous visibility to the “rhel1” window is not necessary.

- In your PuTTY session to “rhel1”, change directory into the largeFG volume. Navigate down into one of the listed directory trees, pick a file that you want to restore, and “corrupt” that file by overwriting it with some content of your own choosing. This example uses the file_00.txt file located under the volume’s dir_0/subdir_000 directory, but you can pick a different file under /svm1/largeFG if you wish.

```
[root@rhell1 flexgroups]# cd /svm1/largeFG
[root@rhell1 largeFG]# ls
dir_0  dir_1  dir_2  dir_3  largedir_0  largedir_1
[root@rhell1 largeFG]# cd dir_0/subdir_000
[root@rhell1 subdir_000]# ls
file_00.txt  file_05.txt  file_10.txt  file_15.txt  file_20.txt
file_01.txt  file_06.txt  file_11.txt  file_16.txt  file_21.txt
file_02.txt  file_07.txt  file_12.txt  file_17.txt  file_22.txt
file_03.txt  file_08.txt  file_13.txt  file_18.txt  file_23.txt
file_04.txt  file_09.txt  file_14.txt  file_19.txt  file_24.txt
[root@rhell1 subdir_000]# cat file_00.txt
iuOm4MHY8JCEHoro2BK1CwIMwd7duthcQ
[root@rhell1 subdir_000]# echo junk text > file_00.txt
[root@rhell1 subdir_000]# cat file_00.txt
junk text
[root@rhell1 subdir_000]#
```

- In your PuTTY session for “cluster1”, set your CLI session privilege level to “admin”, the default privilege level for a CLI session. query for the snapshot policy that the largeFG volume is using, and then list the snapshot retention details for that policy.

```
cluster1::*> set -priv admin

cluster1::> volume show -vserver svml -volume largeFG -fields snapshot-policy
vserver volume  snapshot-policy
-----
svml          largeFG default

cluster1::> snapshot policy show -policy default
Vserver: cluster1

Policy Name          Number of Is
Schedules Enabled Comment
-----
default              3 true      Default policy with hourly, daily & weekly schedules.
Schedule      Count Prefix      SnapMirror Label      Retention Period
-----
hourly         6 hourly      -                  0 seconds
daily          2 daily        daily              0 seconds
weekly         2 weekly       weekly             0 seconds

cluster1::>
```

The “default” policy retains six hourly snapshots, two daily snapshots, and two weekly snapshots.

- List the snapshots currently available for the “largeFG” volume. Remember, this is the source volume for the data protection relationship.

```
cluster1::> snap list -vserver svml -volume largeFG

Vserver Volume  Snapshot
-----
svml      largeFG
base
daily.2023-04-28_0010
daily.2023-04-29_0010
hourly.2023-05-13_1705
hourly.2023-05-13_1805
hourly.2023-05-13_1905
hourly.2023-05-13_2005
hourly.2023-05-13_2105
hourly.2023-05-13_2205
snapmirror.e6a71223-aef8-11ed-8c75-005056b0ebc2_2162646627.2023-05-13_220503

10 entries were displayed.

cluster1::>
```

- The “base” snapshot was created shortly after “largeFG” was populated with its file dataset.
- The daily.*, hourly.*, and weekly.* (if any) snapshots are created by the “default” snapshot protection you examined in the preceding step. The specific list of these you see in your lab will vary depending on how long your lab session has been running.
- The “snapmirror.*” snapshot updates as part of every new snapmirror update operation.

- In your PuTTY session for “cluster2”, set your CLI session privilege level to “admin”, then list the current SnapMirror data protection relationships.

```
cluster2::*> set -priv admin

cluster2::> snapmirror show

Source      Destination Mirror Relationship Total Progress
Path        Type Path      State Status      Progress Healthy Last Updated
-----
svml:largeFG XDP svml-dst:vol_largeFG_dest Snapmirrored Idle - true -
```

```
cluster2::>
```

The cluster has a single SnapMirror relationship, protecting the largeFG volume on the svm1 storage virtual machine (hosted on cluster1) to the vol_largeFG_dest volume on the svm1-dest storage virtual machine (hosted on this cluster, cluster2).

8. List the name of the SnapMirror policy used by this replication relationship.

```
cluster2::> snapmirror show -fields policy
source-path  destination-path  policy
-----
svm1:largeFG svm1-dst:vol_largeFG_dest Asynchronous

cluster2::>
```

9. List the details of the “Asynchronous” SnapMirror policy.

```
cluster2::> snapmirror policy show -policy Asynchronous
Vserver Policy          Policy Number      Transfer
Name      Name              Type    Of Rules  Tries  Priority  Comment
-----
cluster2
    Asynchronous          mirror-vault    3        8    normal  A unified Asynchronous
    SnapMirror and SnapVault policy for mirroring the latest active file system and
    daily and weekly Snapshot copies with an hourly transfer schedule.
    SnapMirror Label: sm_created                      Keep:      1
                      daily                          7
                      weekly                        52
                                           Total Keep: 60

cluster2::>
```

The “Asynchronous” policy directs SnapMirror to replicate any snapshots it finds on the source volume that include the labels “sm_created”, “daily”, or “weekly” to the destination volume. The destination volume will retain the seven most recent daily snapshots and the 52 most recent weekly snapshots.

10. List the snapshots currently available the data protection relationship’s destination volume, vol_largeFG_dest, and then choose one to use for your upcoming file restore operation.

```
cluster2::> snapshot list -vserver svm1-dst -volume vol_largeFG_dest
Vserver Volume Snapshot                                     Size Total% Used%
-----
svm1-dst vol_largeFG dest
    daily.2023-04-28_0010                                16.19GB    1%   50%
    daily.2023-04-29_0010                                16.19GB    1%   50%
    snapmirror.e6a71223-aef8-11ed-8c75-005056b0ebc2_2162646627.2023-05-13_220503
                                                16.20GB    1%   50%
    snapmirror.e6a71223-aef8-11ed-8c75-005056b0ebc2_2162646627.2023-05-13_230503
                                                16.20GB    1%   50%
4 entries were displayed.

cluster2::>
```

If your lab contains “daily” or “weekly” snapshots that are present on “vol_largeFG_dest” but not present on “largeFG” then feel free to pick that particular snapshot name as your source snapshot. Otherwise pick one of the older listed daily or weekly snapshots to use. The only reason you don’t want to pick the most recent daily or weekly snapshot is that there exists a very small chance that ONTAP might have created and replicated a new such snapshot after you had already overwritten the file you chose to “corrupt” in step 4.

11. In your PuTTY session for “cluster1”, restore your selected file from your selected snapshot. This example specifies the snapshot “daily.2023-04-29_0010” and the file “/dir_0/subdir_000/file_00.txt”, but the values you need to use in your lab may be different. The argument to the “--file-list” option specifies the name of the source and destination files, separated by “,@”. By specifying the same file in both positions, you are directing ONTAP to restore the file to its original location. If you wanted to restore the file under a different file name then you would instead list that new file name in the second position.

```
cluster1::> snapmirror restore -destination-path svm1:largeFG -source-path svm1-dst:vol_largeFG_dest
-source-snapshot daily.2023-04-29_0010 -file-list /dir_0/subdir_000/file_00.txt,@/dir_0/subdir_000/
file_00.txt
```

```
[Job 192] Job is queued: snapmirror restore from source "svml-dst:vol_largeFG_dest" for the snapshot daily.2023-04-29_0010.
```

```
cluster1::>
```

Takeaway: The procedure to initiate a snapmirror single file restore for a FlexGroup volume is the same as for initiating a snapmirror single file restore of a FlexVol volume. In both cases a single command accomplishes the restore operation.

12. Periodically query ONTAP for the status of the associated snapmirror restore job, waiting until the restore completes successfully. In this command, specify the job number returned in the preceding step.

```
cluster1::> job show 192
```

Job ID	Name	Owning Vserver	Node	State
197	Snapmirror FG Restore	cluster1	cluster1-01	Running
Description: snapmirror restore from source "svml-dst:vol_largeFG_dest" for the snapshot daily.2023-04-29_0010				

```
cluster1::> cluster1::> job show 192
```

Job ID	Name	Owning Vserver	Node	State
192	Snapmirror FG Restore	cluster1	cluster1-01	Success
Description: snapmirror restore from source "svml-dst:vol_largeFG_dest" for the snapshot daily.2023-04-29_0010				

```
cluster1::>
```

13. In your PuTTY session for "rhel1", verify that your selected file has been restore to its original location and contents that you saw in step 4.

```
[root@rhel1 subdir_000]# pwd
/svml/largeFG/dir_0/subdir_000
[root@rhel1 subdir_000]# cat file_00.txt
iuOm4MHY8JCEHoro2BKlCwIMwd7duthcQ
[root@rhel1 subdir_000]#
```

Takeaway: A storage administrator can quickly restore a file from a SnapMirror destination to the original data protection source FlexGroup volume.

2.7 Convert a FlexVol to a FlexGroup

You can convert an existing FlexVol volume to a FlexGroup volume. The conversion occurs in-place, producing a FlexGroup volume comprised of a single member volume, and the conversion process is non-disruptive to the volume's workload. You might want to undertake a FlexVol to FlexGroup conversion to expand beyond a single FlexVol's limits for capacity and file count, or to leverage a FlexGroup volume's potential for delivering better workload performance.

Converting a FlexVol volume to a FlexGroup volume might not always be the best option. Depending on the version of ONTAP you are running, FlexGroups might not support all of the ONTAP features that you are using for your FlexVol. In addition, ONTAP versions prior to 9.12.1 don't include the FlexGroup rebalance feature, meaning that only a new file ingestion has the potential to place a file on any new constituent volume. This approach requires using a client to move/copy the files from your FlexVol volume to a brand new FlexGroup volume, an inherently disruptive process for the volume you are "converting".

In this lab activity a storage administrator wants to non-disruptively convert an existing FlexVol volume to a FlexGroup volume.

1. This lab activity requires one PuTTY session for “rhel1” and two PuTTY sessions for “cluster1”. If you need to open a new PuTTY session for any of these hosts, on the task bar of Jumphost, right-click the **PuTTY icon** and select **PuTTY** from the context menu.

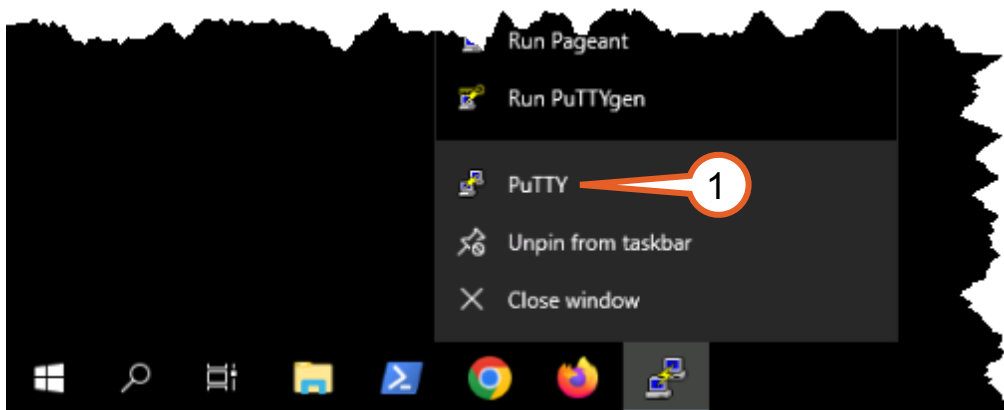


Figure 2-12:

The “PuTTY Configuration” dialog opens.

2. Double-click the session name for **your desired host** in the “Saved Sessions” list.

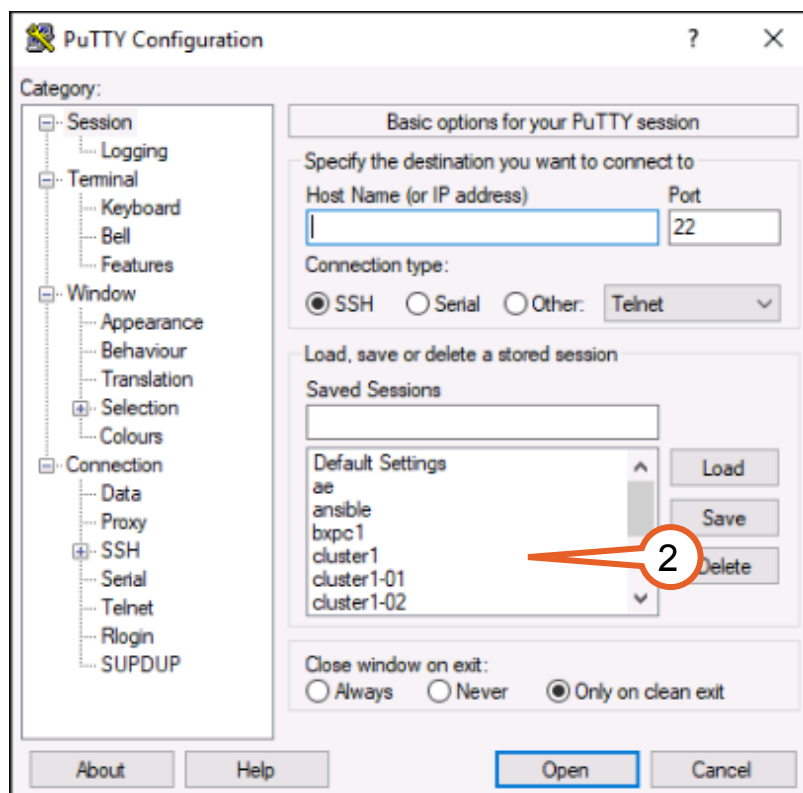


Figure 2-13:

3. Log in using the following credentials:
 - Username (for cluster1): `admin`
 - Username (for rhel1): `root`
 - Password: `Netapp1!`

4. In your first PuTTY session for “cluster1”, set your privilege to the “admin” level and then list the volumes present on the svm1 storage virtual machine.

```
cluster1::> set -priv advanced

Warning: These advanced commands are potentially dangerous; use them only when directed to do so by
NetApp personnel.
Do you want to continue? {y|n}: y

cluster1::*> volume show -vserver svm1
Vserver   Volume      Aggregate      State      Type      Size      Available  Used%
-----
svm1      balanceFG   -              online     RW        20GB      10.40GB    45%
svm1      convertFV   aggr1_cluster1_01
          online     RW        20GB      13.11GB    30%
svm1      demoFG     -              online     RW        200GB     186.4GB    1%
svm1      largeFG    -              online     RW        1.23TB    1.15TB     1%
svm1      smallFG    -              online     RW        11.42GB   3.86GB     64%
svm1      svm1_root  aggr1_cluster1_01
          online     RW        20MB      17.68MB    6%

6 entries were displayed.

cluster1::*>
```

5. List the state of the snapshots for the “convertFV” volume.

```
cluster1::*> snap list -vserver svm1 -volume convertFV -fields state
vserver volume      snapshot      state
-----
svm1      convertFV  daily.2023-04-28_0010 -
svm1      convertFV  daily.2023-04-29_0010 -
svm1      convertFV  hourly.2023-05-13_1905 -
svm1      convertFV  hourly.2023-05-13_2005 -
svm1      convertFV  hourly.2023-05-13_2105 -
svm1      convertFV  hourly.2023-05-13_2205 -
svm1      convertFV  hourly.2023-05-13_2305 -
svm1      convertFV  hourly.2023-05-14_0005 -

8 entries were displayed.

cluster1::*>
```

All of the snapshots report a state of “-”, indicating the snapshot is unrestricted.

6. Prepare this CLI session for later steps by changing its privilege level to “diag”.

```
cluster1::*> set -priv diag

Warning: These diagnostic commands are for use by NetApp personnel only.
Do you want to continue? {y|n}: y

cluster1::*>
```

7. In your PuTTY session for “rhel1”, list the contents of the convertFV volume, then change directory to ~/flexgroups in preparation for upcoming steps in this lab activity.

```
[root@rhel1 subdir_000]# ls /svm1
balanceFG  convertFV  demoFG  largeFG  smallFG
[root@rhel1 subdir_000]# cd /svm1/convertFV
[root@rhel1 convertFV]# ls
largedir_0  largedir_1  largedir_2  largedir_3  largedir_4  largedir_5
[root@rhel1 convertFV]# ls -l largedir_0
total 1028080
-rw-r--r--. 1 root root 104857600 Apr 27 18:27 largefile_0.dat
-rw-r--r--. 1 root root 104857600 Apr 27 18:27 largefile_1.dat
-rw-r--r--. 1 root root 104857600 Apr 27 18:27 largefile_2.dat
-rw-r--r--. 1 root root 104857600 Apr 27 18:27 largefile_3.dat
-rw-r--r--. 1 root root 104857600 Apr 27 18:27 largefile_4.dat
-rw-r--r--. 1 root root 104857600 Apr 27 18:28 largefile_5.dat
-rw-r--r--. 1 root root 104857600 Apr 27 18:28 largefile_6.dat
-rw-r--r--. 1 root root 104857600 Apr 27 18:28 largefile_7.dat
-rw-r--r--. 1 root root 104857600 Apr 27 18:28 largefile_8.dat
```

```
-rw-r--r--. 1 root root 104857600 Apr 27 18:28 largefile_9.dat
[root@rhell convertFV]#
```

This volume contains 6 directories, each containing ten 100 MB files.

8. In your second PuTTY session for “cluster1”, set your privilege level to the “advanced” level and start capturing and outputting periodic cluster-wide I/O statistics for the cluster. Leave the command running for later reference.



Note: This command generates long lines of output that exceed the PuTTY default window widths, producing line-wrapping that can make interpreting the output more difficult. You may want to widen this PuTTY window to eliminate the line wrapping.

```
cluster1::> set -priv advanced

Warning: These advanced commands are potentially dangerous; use them only when
         directed to do so by NetApp personnel.
Do you want to continue? {y|n}: y

cluster1:*> statistics show-periodic
cluster1: cluster.cluster: 5/14/2023 00:01:34
```

cpu		cpu	total	disk		disk	fcache	total	total	data	data	data	cluster
avg	busy	cluster	ops	nfs-ops	cifs-ops	ops	pkts	pkts	sent	busy	recv	sent	busy
recv	sent	read	write	recv	sent	recv	sent	recv	sent	recv	sent	recv	sent
2%	3%	0	0	0	0	0	0	428KB	427KB	0%	734B	613B	0%
427KB	426KB	21.7KB	1.70MB	153	151	578KB	577KB	0%	0B	0B	0B	0B	0%
1%	2%	0	0	0	0	570KB	569KB	0%	0B	0B	0B	0B	0%
578KB	577KB	11.8KB	23.6KB	291	291	570KB	569KB	0%	0B	0B	0B	0B	0%
1%	2%	0	0	0	0	570KB	569KB	0%	0B	0B	0B	0B	0%
570KB	569KB	11.9KB	35.6KB	193	193								

9. To achieve the best experience for the remainder of this lab activity, arrange the “rhel1” and two “cluster1” PuTTY sessions on your display so that you can see the contents of all three windows simultaneously.
10. In your PuTTY session for “rhel1”, initiate a workload against the convertFV volume. The populate_convertFV.sh script will overwrite the volume’s existing set of files with completely new random data. The script only requires about 90 seconds to complete, so proceed immediately to the next step, without waiting for this script to finish, so that you can initiate the FlexVol to FlexGroup conversion while the script is still running.

```
[root@rhell convertFV]# cd ~/flexgroups
[root@rhell flexgroups]# ls
balanceFG-binary.json      largeFG-binary.json    populate_largeFG.sh
balanceFG-largebinary.json largeFG-text.json      populate.py
convertFV-binary.json      populate_balanceFG.sh  README.md
demoFG-binary.json         populate_convertFV.sh
demoFG-text.json           populate_demoFG.sh
[root@rhell flexgroups]# ./populate_convertFV.sh

#####
##### Starting File Population for #####
##### /svml/convertFV #####
#####

#####
##### Populating large binary files
#####

Overlaying script defaults using /root/flexgroups/convertFV-binary.json.

Populating /svml/convertFV with the following:
 6 directories (largedir_N).
10 random binary files (largefile_N.dat, 100MiB bytes) per directory.
100% of all created files will be unique.
Files that already exist will be overwritten.
In total, up to 60 files will potentially be distributed across 7 directories.

Population task starting now --->
..
```

11. In your second PuTTY session for “cluster1”, the one that is displaying the I/O statistics, observe that after the populate script started the cluster’s reported “total ops” and “total nfs-ops” values have changed from zeroes to as high as 3000 ops per second. The cluster’s CPU utilization has also risen from a low single digit percentage to much higher value.

 **Note:** There is no command to run in this step.

```
.
.
.
40% 61%      0      0      0      0      12.5MB 12.5MB 0%      692B      472B      10%
12.5MB 12.5MB 417KB 50.3MB 2710 2707      568KB 567KB 0%      0B      0B      0%
27% 36%      0      0      0      0      150
567KB 566KB 31.7KB 17.8KB 153
cluster1: cluster.cluster: 5/14/2023 00:15:51
cpu cpu      total      disk      disk      fcache      total      total data      data      data cluster
cluster cluster ops  nfs-ops cifs-ops pkts      pkts      sent busy      recv      sent busy
avg busy      ops  read  write  recv  ops  recv      sent busy      recv      sent busy
recv      sent      read  write  recv  ops  sent
-----
2% 3%      0      0      0      0      364KB 360KB 0%      3.95KB 1.69KB 0%
359KB 358KB 11.9KB 29.7KB 169 150
4% 4%      0      0      0      0      443KB 442KB 0%      0B      0B      0%
443KB 442KB 267KB 12.6MB 251 249
3% 5%      0      0      0      0      420KB 420KB 0%      67B      64B      0%
420KB 420KB 15.6KB 24.7MB 135 135
1% 2%      0      0      0      0      578KB 577KB 0%      0B      0B      0%
578KB 577KB 11.8KB 17.7KB 273 273
23% 31% 1276 1276 0 0 66.0MB 15.5MB 0% 4.05KB 313B 21%
50.5MB 15.1MB 7.13MB 72.7MB 18566 11292
48% 51% 1571 1571 0 0 139MB 75.3MB 0% 0B 0B 35%
51.1MB 73.1MB 2.92MB 67.5MB 73578 45631
41% 53% 2445 2445 0 0 142MB 96.7MB 0% 3.22KB 1.12KB 44%
57.9MB 94.6MB 3.24MB 93.7MB 73253 46773
58% 76% 1797 1797 0 0 159MB 69.0MB 0% 114B 77B 37%
91.8MB 67.3MB 3.97MB 81.8MB 64381 40408
71% 74% 2704 2704 0 0 152MB 76.6MB 0% 2.63KB 2.06KB 31%
75.3MB 74.6MB 3.94MB 94.0MB 69263 43156
74% 80% 1519 1519 0 0 144MB 74.1MB 0% 615B 693B 30%
71.7MB 72.2MB 2.54MB 96.8MB 65627 41337
.
.
.
```

12. In your first PuTTY session window for “cluster1”, the one not outputting the I/O statistics, initiate the FlexVol to FlexGroup conversion for convertFV. The command prompt will not return until the conversion completes, which usually takes less than a minute.

```
cluster1::*> volume conversion start -vserver svml -volume convertFV

Warning: * After the volume is converted to a FlexGroup, it will not be possible to change it back
to a flexible
volume.
* Converting flexible volume "convertFV" in Vserver "svml" to a FlexGroup will cause the
state of all
Snapshot copies from the volume to be set to "pre-conversion". Pre-conversion Snapshot
copies cannot be
restored.
* It is recommended that the size of volume "convertFV" in Vserver "svml" be at least
"100GB" for
optimal performance.
* Converting the volume to a FlexGroup will not add additional resources for capacity.
After converting,
use the "volume expand" command to add resources.
Do you want to continue? {y\n}: y
[Job 199] Job succeeded: success

cluster1::*>
```

13. In your PuTTY session to “rhel1”, if the populate_convertFV.sh script is still running, wait for that script to complete, then review the script output to confirm that it reported no I/O errors, confirming that the conversion operation was non-disruptive.



Note: You do not actually run the command in this step, you just examine the output of the command you ran in step 8.

```
[root@rhel1 flexgroups]# ./populate_convertFV.sh

#####
##### Starting File Population for #####
##### /svml/convertFV #####
#####

#####
##### Populating large binary files
#####

Overlaying script defaults using /root/flexgroups/convertFV-binary.json.

Populating /svml/convertFV with the following:
  6 directories (largedir N).
  10 random binary files (largefile_N.dat, 100MiB bytes) per directory.
  100% of all created files will be unique.
  Files that already exist will be overwritten.
  In total, up to 60 files will potentially be distributed across 7 directories.

Population task starting now --->
.....
<--- Population task complete

Created/updated 60 out of 60 potential file candidates
Population task elapsed time = 86.55 seconds

#####
##### File Population Complete for #####
##### /svml/convertFV #####
##### Total Time = 86 seconds #####
#####

[root@rhel1 flexgroups]#
```

14. In your first PuTTY session to “cluster1”, list the constituent volumes for your now converted “convertFV” FlexGroup volume.

```
cluster1::*> volume show -vserver svml -volume convertFV*
Vserver  Volume      Aggregate      State      Type      Size      Available  Used%
-----
svml     convertFV    -              online     RW         20GB      8.16GB     57%
svml     convertFV__0001
          aggr1_cluster1_01
          online     RW         20GB      8.16GB     57%

2 entries were displayed.

cluster1::*>
```

The “convertFV” volume is now a FlexGroup volume comprised of a single constituent volume. At this point you would typically now add additional constituent volumes to your FlexGroup volume, and then, assuming you are running ONTAP 9.12.1 or later, perhaps perform a FlexGroup rebalance to redistribute some of the files present on the FlexGroup’s original constituent volume to the newly added constituent volumes. However, since you have already seen those procedures in the lab’s earlier activities they are not repeated again here.

15. List the state of the snapshots for the “convertFV” volume.

```
cluster1::*> snap list -vserver svml -volume convertFV -fields state
vserver volume      snapshot              state
-----
svml     convertFV    daily.2023-04-29_0010 pre-conversion
svml     convertFV    hourly.2023-05-13_1905
          pre-conversion
svml     convertFV    hourly.2023-05-13_2005
          pre-conversion
svml     convertFV    hourly.2023-05-13_2105
          pre-conversion
svml     convertFV    hourly.2023-05-13_2205
```

```

svml      convertFV hourly.2023-05-13_2305      pre-conversion
svml      convertFV hourly.2023-05-14_0005      pre-conversion
svml      convertFV daily.2023-05-14_0010      pre-conversion
svml      convertFV weekly.2023-05-14_0015      pre-conversion
svml      convertFV convert.2023-05-14_001614    pre-conversion
10 entries were displayed.
cluster1::*>

```

The snapshots that pre-date your conversion from FlexVol to FlexGroup all now have a state of “pre-conversion”, indicating that ONTAP will not allow you to use these snapshots to perform snapshot restore operations. However, you can still use a storage client to access the contents of these pre-conversion snapshots, as you will see shortly.

Significantly, any new snapshots you create on the FlexGroup volume after conversion operate normally.

16. In your PuTTY session for “rhel1”, change directories into one of the directories on the “convertFV” volume and use this client host to restore one of convertFV’s files from a pre-conversion snapshot you choose from the preceding step.



Important: When selecting which pre-conversion snapshot to use for the following commands, pick one of the daily.* or hourly.* pre-conversion snapshots. The convert.* snapshot was initiated at the start of the FlexVol to FlexGroup conversion process, and that snapshot is likely to include some files that had already been updated by the populate_convertFV.sh you launched in step 10.

```

[root@rhel1 flexgroups]# cd /svml/convertFV
[root@rhel1 convertFV]# ls
largedir_0 largedir_1 largedir_2 largedir_3 largedir_4 largedir_5
[root@rhel1 convertFV]# cd largedir_0
[root@rhel1 largedir_0]# ls
largefile_0.dat largefile_3.dat largefile_6.dat largefile_9.dat
largefile_1.dat largefile_4.dat largefile_7.dat
largefile_2.dat largefile_5.dat largefile_8.dat
[root@rhel1 largedir_0]# ls .snapshot
convert.2023-05-14_001614 hourly.2023-05-13_2005 hourly.2023-05-14_0005
daily.2023-04-29_0010 hourly.2023-05-13_2105 weekly.2023-05-14_0015
daily.2023-05-14_0010 hourly.2023-05-13_2205
hourly.2023-05-13_1905 hourly.2023-05-13_2305
[root@rhel1 largedir_0]# ls .snapshot/daily.2023-04-29_0010
largefile_0.dat largefile_3.dat largefile_6.dat largefile_9.dat
largefile_1.dat largefile_4.dat largefile_7.dat
largefile_2.dat largefile_5.dat largefile_8.dat
[root@rhel1 largedir_0]# diff largefile_0.dat .snapshot/daily.2023-04-29_0010/largefile_0.dat
Binary files largefile_0.dat and .snapshot/daily.2023-04-29_0010/largefile_0.dat differ
[root@rhel1 largedir_0]# cp -p .snapshot/daily.2023-04-29_0010/largefile_0.dat .
cp: overwrite './largefile_0.dat'? y
[root@rhel1 largedir_0]# ls -l
total 1028080
-rw-r--r--. 1 root root 104857600 Apr 27 18:27 largefile_0.dat
-rw-r--r--. 1 root root 104857600 May 14 00:16 largefile_1.dat
-rw-r--r--. 1 root root 104857600 May 14 00:16 largefile_2.dat
-rw-r--r--. 1 root root 104857600 May 14 00:16 largefile_3.dat
-rw-r--r--. 1 root root 104857600 May 14 00:16 largefile_4.dat
-rw-r--r--. 1 root root 104857600 May 14 00:16 largefile_5.dat
-rw-r--r--. 1 root root 104857600 May 14 00:16 largefile_6.dat
-rw-r--r--. 1 root root 104857600 May 14 00:16 largefile_7.dat
-rw-r--r--. 1 root root 104857600 May 14 00:16 largefile_8.dat
-rw-r--r--. 1 root root 104857600 May 14 00:16 largefile_9.dat
[root@rhel1 largedir_0]#

```

17. In your second PuTTY session to “cluster1”, with that window in focus, terminate the running “statistics show-periodic” by pressing `ctrl-c`. If you are running a Mac, you may need to press `cmd-c` instead.

```

.
.
.
53% 62% 2124 2124 0 0 155MB 86.2MB 0% 1.19KB 296B 34%
68.0MB 84.0MB 4.47MB 101MB 76137 47357

```

```

34% 38% 1830 1830 0 0 151MB 69.1MB 0% 2.60KB 1.51KB 34%
83.7MB 67.3MB 3.73MB 86.5MB 63077 39713
31% 39% 2263 2263 0 0 139MB 70.7MB 0% 0B 0B 28%
69.0MB 69.0MB 4.26MB 80.3MB 63290 39537
11% 15% 590 590 0 0 60.3MB 30.7MB 0% 0B 0B 12%
30.0MB 29.9MB 1.06MB 38.0MB 27504 17277
2% 2% 0 0 0 0 572KB 571KB 0% 0B 0B 0%
572KB 571KB 17.8KB 29.7KB 194 194
5% 6% 0 0 0 0 300KB 296KB 0% 4.28KB 1.30KB 0%
295KB 295KB 17.8KB 813KB 214 187
3% 4% 0 0 0 0 589KB 587KB 0% 18.6KB 17.9KB 0%
570KB 569KB 17.8KB 11.9KB 234 232
4% 4% 0 0 0 0 584KB 580KB 0% 3.33KB 1.16KB 0%
580KB 579KB 124KB 1.78MB 340 323
3% 4% 0 0 0 0 571KB 570KB 0% 1.46KB 1023B 0%
569KB 569KB 2.42MB 4.03MB 199 192
2% 2% 0 0 0 0 575KB 574KB 0% 864B 871B 0%
574KB 573KB 11.9KB 11.9KB 241 242
cluster1: cluster.cluster: 5/14/2023 00:17:39
cpu cpu total fcache total total data data data cluster
cluster cluster disk disk pkts pkts
avg busy ops nfs-ops cifs-ops ops
recv sent read write recv sent sent busy recv sent busy
-----
Minimums:
1% 2% 0 0 0 0 295KB 294KB 0% 0B 0B 0%
295KB 294KB 0B 5.94KB 112 116
Averages for 113 samples:
26% 33% 750 750 0 0 54.0MB 27.7MB 0% 1.34KB 885B 12%
27.0MB 26.9MB 3.71MB 35.6MB 24529 15456
Maximums:
80% 88% 2998 2998 0 0 210MB 116MB 0% 18.6KB 17.9KB 46%
113MB 113MB 51.1MB 121MB 103940 64802

cluster1::*>

```

Takeaway: Storage administrators can easily and non-disruptively convert existing FlexVol volumes into FlexGroup volumes.

3 Lab Environment

The following diagram illustrates the environment for this lab.

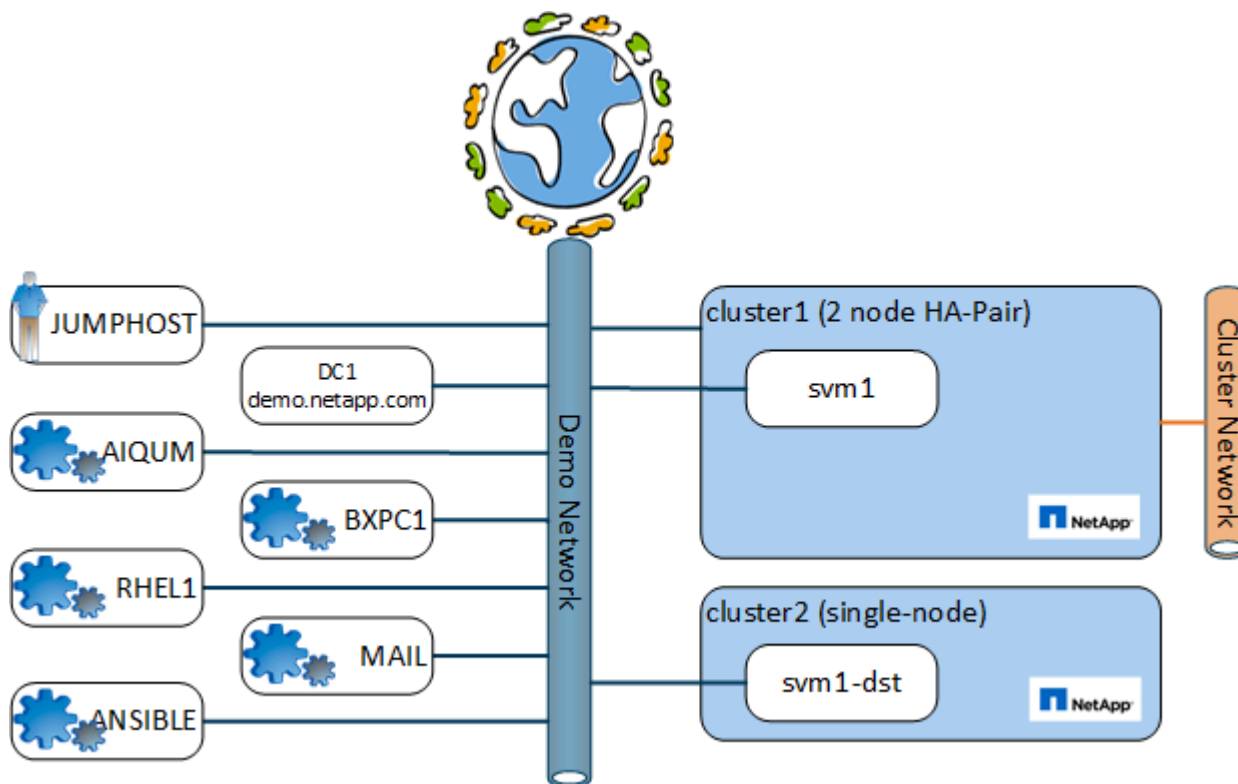


Figure 3-1:

All of the servers and storage controllers presented in this lab are virtual devices, and the networks that interconnect them are exclusive to your lab session. While we encourage you to follow the demonstration steps outlined in this lab guide, you are free to deviate from this guide and experiment with other ONTAP features that interest you. While the virtual storage controllers (vsims) used in this lab offer nearly all of the same functionality as physical storage controllers, they are not capable of providing the same performance as a physical controller and these labs are not suitable for performance testing.

The [Lab Host Credentials](#) table provides a list of the servers and storage controller nodes in the lab, along with their IP address.

Table 1: Lab Host Credentials

Hostname	Description	IP Address(es)	Username	Password
dc1	Windows 2019 Active Directory Server	192.168.0.253	Demo Administrator	Netapp1!
jumphost	Windows 2019 Remote Access host	192.168.0.5	Demo Administrator	Netapp1!
rhel1	RHEL 8.7 x64	192.168.0.61	root	Netapp1!
aiqum	ActiveIQ Unified Manager 9.12	192.168.0.71	admin	Netapp1!
bxpc1	RHEL 8.7 x64	192.168.0.72	root	Netapp1!

Hostname	Description	IP Address(es)	Username	Password
ansible	Ansible 2.9.20 Management Host	192.168.0.188	ansible	Netapp1!
cluster1	ONTAP 9.12.1 cluster	192.168.0.101	admin	Netapp1!
cluster2	ONTAP 9.12.1 cluster	192.168.0.102	admin	Netapp1!

The [Preinstalled NetApp Software](#) table lists the NetApp software that is pre-installed on the various hosts in this lab.

Table 2: Preinstalled NetApp Software

Hostname	Description
Jumphost	NetApp PowerShell Toolkit v4.7.0, Windows Unified Host Utility Kit v7.1.0
aiqum	ActiveIQ Unified Manager 9.12
ansible	netapp-lib (2020.7.16+) for Ansible

 **Note:** This lab does not actually use Ansible. The “ansible” host is only present for lab users who wish to experiment on their own with the combination of Ansible and FlexGroup volumes. For the convenience of those users, Ansible and the netapp-lib library are pre-installed on that host, and the ~ansible/ontap folder contains a copy of the playbooks from the the “Ansible Fundamentals v1.0” lab to help facilitate playbook development. However, those playbooks have not been tested in this lab and are not guaranteed to work here.

4 References

The following documents contain more detail on the NetApp technologies discussed in this lab guide.

- [FlexGroup Volume Management with the CLI](#)
- [TR-4571: NetApp FlexGroup Volume Best Practices and Implementation Guide](#)
- [NetApp ONTAP FlexGroup Volumes Top Best Practices](#)
- [File System Analytics Overview](#)
- [TR-4867: Best Practice Guidelines for ONTAP File System Analytics](#)

5 Version History

Version	Date	Document Version History
1.1.0	May 2023	Initial Release

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